toggle-manhattan

System and Software Architecture Design

Version 0.1

Ongoing

by

Stel Angelidis

Computer Club

October 14, 2026

© Computer Club 2026
ALL RIGHTS RESERVED

Template Attribution

This software architecture design template was developed by **Darian Muresan, Ph.D.**, in conjunction with the AI agents **OpenAI Codex** and **Anthropic Claude Code**.

Contact: dmuresan@stevens.edu
darian@dmmd.net
darian@pictomic.com

toggle-manhattan

System and Software Architecture Design

Version 0.1
Ongoing

Stel Angelidis
Computer Club
October 14, 2026

This document specifies the stakeholder needs, requirements, architecture, implementation constraints, tests, verification evidence, validation evidence, and release criteria for the toggle-manhattan. It is both a printable engineering record and the authoritative input for AI-assisted implementation. A conforming implementation may be regenerated from the complete specification, but no single chapter should be treated as sufficient on its own.

Document History

How to Use This Chapter

Add the newest approved change at the top of the table. Retain every approved entry so that the document's evolution and review history remain auditable. Every history entry shall link to the chapter, section, table, figure, or controlled artifact where the documented change occurred.

Author initials: SAA: Stel Angelidis; **CLAUDE**: Anthropic Claude Code; **DDM**: David Darian Muresan; **CDX**: OpenAI Codex. A comma identifies joint authorship or implementation.

Entries from the template baseline through version 2.7 record the evolution of the master architecture template from which toggle-manhattan was initialized. Project history begins at version 0.1.

Table 2: Document Update History

Date / Version	Author Initials	Change	Changed At
October 14, 2026 / 0.1	SAA, CLAUDE	Initialized toggle-manhattan from template version 2.7: applied the `tgm` file namespace to all project-owned files and TikZ sources; set project name, owner, organization, version, status, and date; recorded the Embedded Systems profile as Applicable on a microcontroller platform; recorded Stel Angelidis as authorized approver; recorded the system purpose, the decision to provide both a physical controller and a phone application, the stakeholders, the privacy, USD 20 component-cost, and microcontroller constraints, and open context items; added the IoT acronym; added a Claude Code repository bootstrap and the initialization prompt record.	Chapter 1, approval authority, profile decisions, system purpose, stakeholders, constraints, Chapter D

Continued on next page

Document Update History (continued)

Date / Version	Author Initials	Change	Changed At
September 18, 2026 / 2.7	DDM, CDX	Required every applicable domain-profile concern to map to a new or existing project requirement, or to an authorized Not Applicable decision; added AI task routing and requirements-work instructions for this translation.	Profile requirement rule, Chapter C, Chapter B
September 18, 2026 / 2.6	DDM, CDX	Added the domain-tailoring framework and Embedded Systems Tailoring Profile; added profile selection to initialization and the Development Plan; and extended AI routing, workflows, prompts, and completion checks for applicable profiles.	Domain profiles, Chapter B, profile decisions, Chapter C
September 9, 2026 / 2.5	DDM, CDX	Required the project-initialization prompt to confirm the repository's AI tools before selecting bootstrap files and to create `SKILL.md` only for a confirmed reusable AI skill.	Project initialization
September 9, 2026 / 2.4	DDM, CDX	Added an explicit copy-and-paste AI request to the project-initialization section, including namespace confirmation, evidence review, gap analysis, human review, controlled bootstrap creation, and build verification.	Project initialization
September 9, 2026 / 2.3	DDM, CDX	Made TikZ the exclusive diagram source format; removed Graphviz generation from both platform build files and removed current fallback, installation, build, troubleshooting, and AI guidance for Graphviz.	TikZ authoring, Chapter A, Chapter C
September 9, 2026 / 2.2	DDM, CDX	Added a concise Introduction workflow for instantiating a project-specific architecture book from the master template, including namespace confirmation, evidence review, dependency-ordered development, AI bootstrap guidance, and build verification.	Project initialization, Chapter C

Continued on next page

Document Update History (continued)

Date / Version	Author Initials	Change	Changed At
September 9, 2026 / 2.1	DDM, CDX	Added the repository AI bootstrap procedure, applicable discovery-file catalog, non-duplication rule, reusable creation prompt, tool-mandated filename exception, and completion check.	AI bootstrap, authoring requirements, example prompts, completion checklist
September 8, 2026 / 2.0	DDM, CDX	Required a human-confirmed project namespace for every new project-owned file; required each TikZ directory to exactly match its including LaTeX file; and renamed both active TikZ sources to comply.	authoring requirements, TikZ authoring, Chapter C, V-model, use-case diagram
September 7, 2026 / 1.9	DDM, CDX	Linked each use-case identifier in the high-level TikZ diagram directly to its authoritative use-case definition.	use-case diagram
September 7, 2026 / 1.8	DDM, CDX	Replaced the active V-model and high-level use-case Graphviz figures with chapter-owned TikZ sources and updated both chapters to include and reference the vector diagrams directly.	V-model, use-case diagram
September 7, 2026 / 1.7	DDM, CDX	Established TikZ as the primary diagram-authoring tool, retained Graphviz as a fallback, documented the source and inclusion pattern, and added the mandatory AI rule for locating and referencing every TikZ source.	Chapter A, Chapter C

Continued on next page

Document Update History (continued)

Date / Version	Author Initials	Change	Changed At
September 6, 2026 / 1.6	DDM, CDX	Added the requirement identifier grammar, active family-code definitions, optional project scope, stable sequence rule, classification boundary, and the controlled AI procedure for proposing and approving a new scope or family without changing existing IDs.	ID convention, Chapter C
August 25, 2026 / 1.5	DDM, CDX	Expanded every formal table to the available portrait or landscape width using flexible multipage columns; retained the existing column hierarchy while increasing space for descriptive content.	History, Chapter 5, Chapter 4, Chapter 14, Chapter 15, Chapter 16, Chapter 17, Chapter C
August 25, 2026 / 1.4	DDM, CDX	Added a dedicated AI Agent Usage Guide with efficient task routing, in-document context packs, workflows, checklists, human-review boundaries, and reusable prompts; added a concise AI entry point in the Introduction; and consolidated duplicated guidance.	Chapter C, Chapter 1, Chapter A
August 25, 2026 / 1.3	DDM, CDX	Reorganized the requirement catalog into explicit families and subtypes; classified scalability, reliability, security, and related characteristics as quality attributes; and clarified the role of specialized working registers.	Catalog, Chapter 5
August 25, 2026 / 1.2	DDM, CDX	Added the Open, In Progress, Done, and Retired requirement lifecycle; defined its relationship to priority and implementation version; and added Status to every requirement register.	Status, Chapter 5

Continued on next page

Document Update History (continued)

Date / Version	Author Initials	Change	Changed At
August 25, 2026 / 1.1	DDM, CDX	Split requirements into category-specific registers; added business requirements, a broad category catalog, functional-versus-quality guidance, and instructions for project-specific categories; and completed reciprocal mappings.	B, F, Q, I, S, D, C, Catalog
August 25, 2026 / 1.0	DDM, CDX	Added explicit generic interface, security, data, and constraint requirement examples with reciprocal story, use-case, test, traceability, and verification mappings.	Chapter 5, US, UC, TC, RTM, VER
August 25, 2026 / 0.9	DDM, CDX	Removed all end-input boundaries and obsolete trailing project content; retained only generic template material; and added an AI-assisted architecture workflow with human-review and prompt-log guidance.	Chapter 1, AI workflow, Chapter D
August 25, 2026 / 0.8	DDM, CDX	Changed the joint-author separator from a slash to a comma.	History
August 25, 2026 / 0.7	DDM, CDX	Replaced template-originator initials with named author initials and documented Codex joint authorship and implementation.	History
August 25, 2026 / 0.6	DDM, CDX	Added visible template attribution, author contact information, AI-agent development credit, PDF creator metadata, source-header provenance, and authoring-guide provenance guidance.	Template attribution, Chapter A
August 25, 2026 / 0.5	DDM, CDX	Replaced the literal x in the requirements implementation-version field with the empty-set symbol to identify requirements with no assigned implementation version.	Chapter 5
August 25, 2026 / 0.4	DDM, CDX	Expanded the generic introduction with the central lesson, source-of-truth guidance, and V-model; added a generic high-level use-case diagram; and expanded the authoring and tool-installation appendix.	Chapter 1, V-model, Chapter 6, Use-case diagram, Chapter A

Continued on next page

Document Update History (continued)

Date / Version	Author Initials	Change	Changed At
August 25, 2026 / 0.3	DDM, CDX	Added linked history records; glossary-linked priority terms; user-story, traceability, verification, and validation registers; implementation-version status; and a final index chapter.	History, Chapter 1, Chapter 5, Traceability, Verification, Validation, Index
August 24, 2026 / 0.2	DDM, CDX	Added longtable history and requirements registers, priority and implemented-version fields, reciprocal use-case mappings, and test-case mappings.	History, Chapter 5, Chapter 6, Chapter 14
[baseline date / 0.1]	DDM	Generic software architecture design template baseline.	Chapter 1

Table of Contents

Template Attribution iii

Document History v

1 Introduction 1
- 1.1 How to Use This Chapter . . . 1
- 1.2 Purpose . . . 1
- 1.3 Scope . . . 1
- 1.4 Audience . . . 2
- 1.5 Normative Language and Priority . . . 2
- 1.6 Document as Source of Truth . . . 2
- 1.7 Initializing a New Project from This Template . . . 2
- 1.8 Template Tailoring and Domain Profiles . . . 3
- 1.9 Using This Template with AI Agents . . . 4
- 1.10 V-Model Organization . . . 4

2 Development Plan 6
- 2.1 How to Use This Chapter . . . 6
- 2.2 Roles and Approval Authority . . . 6
- 2.3 Domain Profile Selection . . . 6
- 2.4 Example . . . 7

3 System Context 8
- 3.1 How to Use This Chapter . . . 8
- 3.2 System Purpose . . . 8
- 3.3 Stakeholders . . . 8
- 3.4 Stakeholder Constraints . . . 8
- 3.5 Open Context Items . . . 9

4 User Stories 10
- 4.1 How to Use This Chapter . . . 10
- 4.2 User Story Register . . . 10
- 4.3 Review Rules . . . 11

5 Requirements **12**
5.1 How to Use This Chapter . . . 12
5.2 Requirement Identifier Naming Convention . . . 12
5.3 Functional and Quality Requirements . . . 14
5.4 Requirement Status Model . . . 14
5.5 Requirement Category Catalog . . . 15
5.6 Domain Profile Impact on Requirements . . . 15
5.7 Creating a Project-Specific Category . . . 18
5.8 Review Rules . . . 19
5.9 Category-Specific Requirement Registers . . . 20
5.9.1 Business Requirements . . . 20
5.9.2 Functional Requirements . . . 21
5.9.3 Quality Requirements . . . 21
5.9.4 Interface Requirements . . . 22
5.9.5 Security Requirements . . . 22
5.9.6 Data Requirements . . . 23
5.9.7 Constraint Requirements . . . 23

6 Use Cases **24**
6.1 How to Use This Chapter . . . 24
6.2 High-Level Use Case Diagram . . . 24
6.3 Submit a Service Request . . . 24
6.4 View Request Status . . . 25
6.5 Export Request History . . . 25

7 Architecture Overview **27**
7.1 How to Use This Chapter . . . 27
7.2 Decision Template and Example . . . 27

8 User Interface Design **28**
8.1 How to Use This Chapter . . . 28
8.2 Example . . . 28

9 Logical View **29**
9.1 How to Use This Chapter . . . 29
9.2 Example . . . 29

10 Process View **30**
10.1 How to Use This Chapter . . . 30
10.2 Example State Model . . . 30

11 Development View **31**
11.1 How to Use This Chapter . . . 31
11.2 Example . . . 31

12 Physical View **32**
12.1 How to Use This Chapter . . . 32
12.2 Example . . . 32

13 Implementation **33**
13.1 How to Use This Chapter . . . 33
13.2 Example . . . 33

14 Tests **34**
14.1 How to Use This Chapter . . . 34
14.2 Example Test Cases . . . 34

15 Traceability **36**
15.1 How to Use This Chapter . . . 36
15.2 Traceability Register . . . 36

16 Verification **38**
16.1 How to Use This Chapter . . . 38
16.2 Verification Register . . . 38

17 Validation **40**
17.1 How to Use This Chapter . . . 40
17.2 Validation Register . . . 40

18 Release **42**
18.1 How to Use This Chapter . . . 42
18.2 Example Checklist . . . 42

Bibliography **43**

A Document Authoring Guide **44**
A.1 How to Use This Appendix . . . 44
A.2 Template Provenance . . . 44
A.3 AI-Assisted Authoring . . . 44
A.4 File Naming . . . 44
A.4.1 Document-Authoring Requirements . . . 45
A.5 One Responsibility per Chapter . . . 46
A.6 Adding a Chapter . . . 46
A.7 Diagram Authoring with TikZ . . . 46
A.8 Building the PDF . . . 47
A.8.1 Installing the Tools on macOS . . . 47
A.8.2 Installing the Tools on Windows . . . 48
A.8.3 Installing the Tools on Ubuntu or Debian Linux . . . 48
A.8.4 AI-Agent Installation . . . 49
A.8.5 Building on macOS and Linux . . . 49

A.8.6 Building with Windows PowerShell . . . 49
A.8.7 Manual Build Sequence . . . 50
A.8.8 Build Verification and Troubleshooting . . . 50
A.9 Authoring Rules . . . 50
A.10 Citations and Bibliography . . . 51
A.11 Selective Builds . . . 51

B Embedded Systems Tailoring Profile **52**
B.1 How to Use This Appendix . . . 52
B.2 Applicability . . . 52
B.3 Core-Chapter Tailoring Map . . . 53
B.4 Hardware–Software Boundary and Allocation . . . 54
B.5 Embedded Requirement Categories . . . 54
B.6 Required Architecture Views . . . 55
B.7 Interfaces and Protocols . . . 56
B.8 Resource Budgets . . . 56
B.9 Safety, Reliability, and Fault Management . . . 57
B.10 Embedded Security . . . 57
B.11 Toolchain and Configuration Control . . . 58
B.12 Embedded Verification and Validation . . . 58
B.13 AI Tailoring Instructions . . . 58
B.14 Embedded-Profile Completion Checklist . . . 59

C AI Agent Usage Guide **60**
C.1 How to Use This Appendix . . . 60
C.2 Governing Principles . . . 60
C.3 AI Quick Start . . . 60
C.4 Repository AI Bootstrap . . . 61
C.5 Task Routing . . . 62
C.6 Task-Specific Context Packs . . . 63
C.6.1 Domain-Profile Context . . . 64
C.6.2 Requirements Context . . . 64
C.6.3 Use-Case Context . . . 64
C.6.4 Architecture Context . . . 64
C.6.5 Testing and Evidence Context . . . 64
C.6.6 Release Context . . . 65
C.6.7 Document-Authoring Context . . . 65
C.7 Mandatory Consistency Rules . . . 65
C.8 Standard AI Workflows . . . 66
C.8.1 Initialize a Project . . . 66
C.8.2 Add or Change a Requirement . . . 67
C.8.3 Add or Change a Use Case . . . 67
C.8.4 Develop or Change Architecture . . . 67
C.8.5 Add Tests or Evidence . . . 67

C.8.6 Review and Prepare a Release . . . 67
C.9 Example AI Requests and Prompts . . . 67
C.9.1 Initialize the Architecture Book . . . 67
C.9.2 Tailor an Embedded System . . . 68
C.9.3 Create Repository AI Bootstrap Files . . . 68
C.9.4 Develop Requirements . . . 68
C.9.5 Develop Use Cases . . . 69
C.9.6 Develop an Architecture Decision . . . 70
C.9.7 Develop Tests and Traceability . . . 70
C.9.8 Review Release Readiness . . . 70
C.10 Human Decisions and Information Boundaries . . . 70
C.11 Completion Checklist . . . 70

D AI Prompt Log **72**
D.1 How to Use This Appendix . . . 72
D.2 Example Record . . . 72
D.3 Requirement Scope or Family Extension Record . . . 72
D.4 Project Initialization Record . . . 72

E Test Evidence **74**
E.1 How to Use This Appendix . . . 74
E.2 Example Evidence Record . . . 74

Glossary **76**

Acronyms **77**

Index **78**

List of Tables

2 Document Update History v

1.1 Domain Profile Applicability 3

2.1 Project Domain Profile Decisions 6

4.1 User Story Register 10

5.1 Requirement Identifier Segments and Family Codes 12
5.2 Requirement Status Definitions 14
5.3 Requirement Families, Subtypes, and Quality Attributes 16
5.4 Business Requirements Register 20
5.5 Functional Requirements Register 21
5.6 Quality Requirements Register 21
5.7 Interface Requirements Register 22
5.8 Security Requirements Register 22
5.9 Data Requirements Register 23
5.10 Constraint Requirements Register 23

14.1 Test Case Examples 34

15.1 Traceability Register 37

16.1 Verification Register 38

17.1 Validation Register 40

A.1 Document-Authoring Requirements 45

B.1 Embedded-Systems Core-Chapter Tailoring Map 53
B.2 Embedded-System Requirement Categories 54
B.3 Candidate Embedded-System Architecture Views 55
B.4 Generic Embedded Resource Budget 57

C.1 Repository AI Bootstrap Files 61
C.2 AI Task-Routing Table 62

C.3 AI-Assisted Change Completion Checklist 71

List of Figures

1.1 V-model relationships used by this specification. 5

6.1 Generic high-level use case diagram. 24

Chapter 1

Introduction

1.1 How to Use This Chapter

State the document purpose, scope, audience, conventions, and relationship to other project artifacts. Replace bracketed text with project-specific content. Keep binding obligations in the Requirements chapter.

1.2 Purpose

This document is the authoritative engineering specification for toggle-manhattan. It records why the system is needed, what it shall do, how it is structured, how it will be tested, and what evidence is required before release. It is readable by people and structured for use by engineering and artificial intelligence (AI)-assisted development tools.

Central lesson

A vague request produces unpredictable software. A precise, traceable specification allows engineers to generate, review, test, and, when necessary, regenerate a conforming implementation.

1.3 Scope

Define the system boundary, included capabilities, exclusions, operating environment, major constraints, and the release covered by this document.

toggle-manhattan is a system that controls the opening and closing of a curtain. The current decision is that a user commands the curtain through both a physical controller and a phone application. Stakeholders are identified in the System Context. Included capabilities, exclusions, target environment, external dependencies, and the release covered by this document are TBD and require supporting project evidence. Open context items are listed in Chapter 3.

1.4 Audience

Identify the people who create, approve, implement, test, operate, maintain, audit, or use the system. State which sections are most relevant to each role.

Generic example: The primary audiences are customer representatives, systems engineers, architects, developers, testers, operators, maintainers, security reviewers, and project approvers. One person may perform several roles on a small project.

1.5 Normative Language and Priority

Use **shall** for every binding requirement. Do not use **should** as guidance in normative statements. Assign each requirement one priority: **Must**, **Should**, **Could**, or **Would**. Priority expresses delivery importance; it does not change whether a statement containing **shall** is binding for its assigned implementation version.

1.6 Document as Source of Truth

Requirements alone do not define a reproducible implementation. The complete source of truth consists of stakeholder needs, user stories, requirements, use cases, architecture and interface decisions, implementation constraints, tests, verification evidence, validation evidence, and release criteria. Generated code is an implementation artifact governed by this specification.

1.7 Initializing a New Project from This Template

Create a project-specific architecture book from a copy of this template; do not place project-specific content in the master template. Before creating or renaming project-owned files, confirm the project name, system boundary, repository location, document owner, and project file namespace. Preserve tool-mandated discovery filenames, such as `AGENTS.md`, when an AI tool requires them.

Gather the approved project evidence and identify missing scope, stakeholders, interfaces, constraints, quality attributes, risks, verification evidence, validation evidence, and approvals before replacing placeholders. Replace only content supported by that evidence. Update the document metadata, title material, contact information, and document history; then develop stakeholder needs, user stories, requirements, use cases, architecture, tests, traceability, verification, validation, and release content in dependency order. Preserve stable identifiers and reciprocal mappings throughout the document.

Human authors and AI agents shall follow the detailed Initialize a Project workflow, repository-bootstrap procedure, naming rules, and completion checklist in Chapter C. After initialization, compile the complete document, review the build log, and visually inspect the generated PDF. The master template remains the reusable reference for later projects.

To initialize a project with an AI agent, replace the bracketed project name and give the agent the following request:

```
Read Section 1.7, "Initializing a New Project from This Template,"
and the authoritative AI Agent Usage Guide in tgmAIAgentGuide.tex.
Initialize this copy of the architecture template for [Project Name].
Before editing, ask me to confirm the project file namespace, which
AI tools this repository will use, and which domain tailoring profiles
apply. Review the repository and supplied project evidence, then
produce a gap list of missing scope,
stakeholders, interfaces, constraints, quality attributes, risks,
evidence, and approvals. Do not invent project facts or replace
unsupported placeholders. After I review the gap list, follow the
documented initialization workflow and dependency order. Create only
the applicable AI bootstrap files. Create SKILL.md only if I confirm
that the repository is intended to provide a reusable AI skill. Update
document history, compile the complete document, inspect the build
log, and visually inspect the resulting PDF. Report unresolved
decisions that require human approval.
```

1.8 Template Tailoring and Domain Profiles

The core chapters form a generic software-architecture template. A domain tailoring profile augments those chapters with concerns, artifacts, diagrams, and review criteria for a particular class of system. It does not replace a core chapter or create a second authoritative location for project facts. Requirements, interfaces, architecture decisions, tests, and evidence remain in their normal core chapters and link to one another through their stable IDs.

Select applicable profiles during project initialization. More than one profile may apply. Record each applicability decision and rationale in Chapter 2, retain applicable profile appendices as authoritative tailoring guidance, and exclude an inapplicable profile from an instantiated project only after recording the decision. If profile guidance conflicts, identify the conflict and obtain an authorized human decision before editing controlled project content.

Table 1.1: Domain Profile Applicability

Profile	**Applicability**	**Decision and Rationale**
Embedded Systems	Applicable	The system controls the physical movement of a curtain. Decision record: project profile decisions.
[Additional profile]	*[Applicable / Not Applicable / TBD]*	*[Reason and relationship to other profiles]*

When the Embedded Systems profile is Applicable or TBD, read Chapter B during initialization and for every task affected by hardware, firmware, real-time behavior, constrained resources, physical interfaces, safety, security, manufacturing, or field service.

1.9 Using This Template with AI Agents

AI agents such as OpenAI Codex and Anthropic Claude Code can use the LaTeX sources as structured project context. The agent shall begin with Chapter C, select the applicable task-routing entry, and read only the authoritative files identified for that task. It shall expand the context only when a referenced dependency or unresolved inconsistency requires it. This keeps routine work focused without weakening traceability.

The AI Agent Usage Guide contains the task-routing table, context packs, mandatory rules, standard workflows, completion checklists, human-review boundaries, and reusable prompts. Record material AI-assisted decisions in Chapter D.

The following minimal request starts an AI-assisted task:

```
Read tgmAIAgentGuide.tex first. Identify the applicable task and
load only the files listed for it. Follow the template's stable-ID,
priority, status, traceability, history, build, and review rules.
Use only supplied project evidence. Report missing information and
decisions that require human approval. Complete the requested task,
compile the document, and visually inspect the affected PDF pages.
```

1.10 V-Model Organization

Figure 1.1 summarizes the V-model organization used by this template. The left side moves from stakeholder needs toward detailed design. The bottom represents implementation. The right side integrates the system and gathers evidence, pairing acceptance with stakeholder needs, system verification with requirements, integration tests with architecture, and unit tests with detailed design.

This organization is compatible with the system and software life-cycle process frameworks in [1, 2]; those standards do not require a particular life-cycle model.

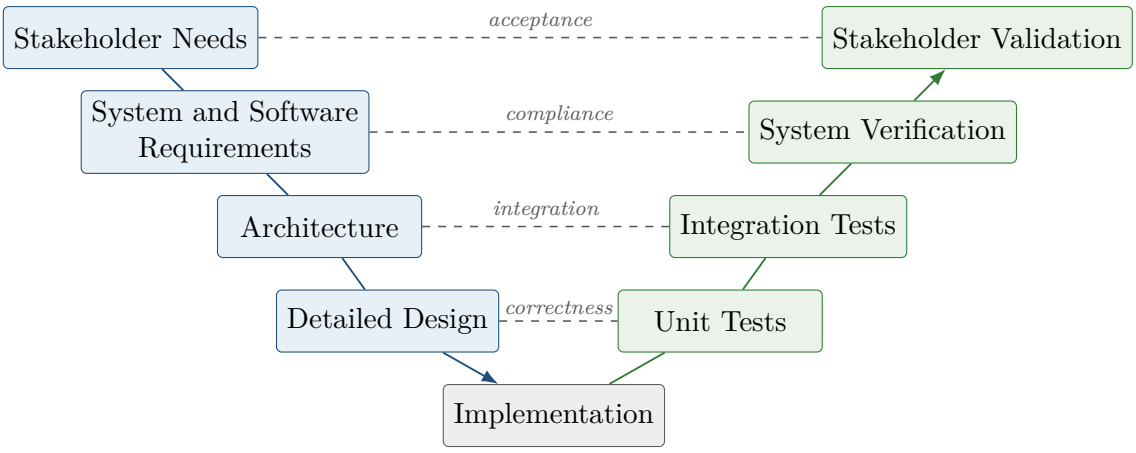


Figure 1.1: V-model relationships used by this specification.

Chapter 2

Development Plan

2.1 How to Use This Chapter

Describe governance, roles, milestones, change control, and risks. Adapt the plan to the lifecycle the team will actually follow.

2.2 Roles and Approval Authority

Stel Angelidis is the document owner and the authorized approver for toggle-manhattan. Other project roles are TBD.

2.3 Domain Profile Selection

Record every domain tailoring profile considered during initialization. An authorized human shall set each profile to Applicable, Not Applicable, or TBD and record the rationale, decision owner, and date. A TBD profile remains in the AI and human review context until resolved. When a profile is Applicable, identify the lifecycle gates, roles, evidence, and reviews it adds to this development plan.

Table 2.1: Project Domain Profile Decisions

Profile	**Status**	**Rationale, Owner, and Date**
Embedded Systems	Applicable	The system controls the physical opening and closing of a curtain. Platform: microcontroller; bare metal or RTOS is TBD. Specific microcontroller, board, and hardware revisions: TBD. Approved by Stel Angelidis on October 14, 2026.
[Additional profile]	*[Applicable / Not Applicable / TBD]*	*[Decision basis, authorized owner, and decision date]*

2.4 Example

The product owner prioritizes outcomes; the architect owns significant design decisions; developers implement approved work; and the test lead retains evidence. Typical gates are needs reviewed, requirements baselined, architecture approved, evidence complete, and release approved. Example risk: an external service may be unavailable; define the failure response and test it before release.

Chapter 3

System Context

3.1 How to Use This Chapter

Define the problem, stakeholders, system boundary, external actors and systems, assumptions, dependencies, and exclusions. Add a context diagram when it clarifies interactions.

3.2 System Purpose

toggle-manhattan controls the opening and closing of a curtain. The current decision is that a user commands the curtain through both a physical controller and a phone application.

3.3 Stakeholders

The stakeholders are people who want a low-cost Internet of Things (IoT) curtain closer that does not invade their privacy. Stel Angelidis is the authorized approver (see approval authority).

3.4 Stakeholder Constraints

The following constraints were stated by the authorized approver on October 14, 2026. They are inputs to the Requirements chapter and are not yet formal requirements.

- **Privacy:** the system collects no data about individual people, and it includes no cameras or microphones.
- **Cost:** the components cost no more than USD 20.
- **Platform:** the system runs on a microcontroller. All other platform characteristics are TBD.

3.5 Open Context Items

The following context is TBD and shall be resolved from project evidence and authorized decisions before requirements are developed:

- the specific users, operators, installers, and maintainers, and any stakeholder groups beyond those identified in Section 3.3;
- the basis of the USD 20 component-cost limit, including whether it applies to the total of all components, which items count (for example the motor, power supply, enclosure, and mounting hardware), the purchase quantity, and the pricing date;
- whether any non-personal operational data, such as curtain position or schedules, may be stored or transmitted, and whether any cloud or third-party service is acceptable;
- the physical-controller and phone-application interfaces, including how the phone application communicates with the system;
- the system boundary, including the curtain mechanism, any actuator, sensors, or power source, and any external network, phone, or cloud service;
- the specific microcontroller, whether it runs bare-metal firmware or an RTOS, the board, and hardware revisions (see Chapter B);
- the physical installation environment, constraints, and assumptions; and
- a context diagram, once the boundary and external actors are known.

Template pattern: [User role] needs to [achieve outcome] because [current problem]. The system manages [owned capability]; [external system] remains outside the boundary and provides [interface]. Record its owner, data exchanged, and failure assumptions.

Chapter 4

User Stories

4.1 How to Use This Chapter

Capture stakeholder goals in accessible language. Stories express value and acceptance intent; precise obligations belong in Requirements. Give each story a stable ID, a priority, and links to one or more requirements that implement it.

4.2 User Story Register

Table 4.1: User Story Register

ID	Story and Acceptance Criteria	Priority	Requirement(s)
US-001	As a customer, I want to submit a service request so that the responsible team can act on it. Acceptance: valid data produces a unique request identifier on every supported client.	Must	REQ-F-001, REQ-C-001
US-002	As a customer, I want to view my request status so that I understand its progress. Acceptance: the current status and most recent update appear within the specified response time, and the transition history is retained.	Should	REQ-F-002, REQ-Q-001, REQ-D-001, REQ-C-001
US-003	As an operations analyst, I want to export request history so that I can perform approved analysis. Acceptance: an authorized actor prepares complete history within five minutes in the approved versioned format without changing source records.	Would	REQ-B-001, REQ-F-003, REQ-I-001, REQ-S-001, REQ-D-001, REQ-C-001

4.3 Review Rules

Confirm that every story identifies a beneficiary, outcome, acceptance criterion, priority, and at least one linked requirement. Maintain reciprocal story links in the traceability register.

Chapter 5

Requirements

5.1 How to Use This Chapter

State singular, necessary, unambiguous, feasible, and verifiable obligations. Every requirement shall have a stable ID, one priority, the version in which it was implemented, at least one mapped use case, and at least one mapped test case. Enter ∅ in **In Version** when no implementation version has been assigned. Preserve IDs when wording or version assignments change.

Select the requirement categories that make the specification easier to review and govern. Keep each requirement in one primary register and use traceability links instead of copying the same requirement into multiple categories.

5.2 Requirement Identifier Naming Convention

Requirement identifiers use the following grammar:

`REQ-[<scope>-]<family>-<sequence>`

The optional scope qualifier identifies a project-defined subsystem, product, content type, or other governed area. Omit it for project-wide requirements. The sequence is a three-digit, zero-padded number that is unique within the exact scope and family combination. For example, `REQ-F-004` is the fourth project-wide functional requirement, while `REQ-API-Q-003` is the third quality requirement in a project-defined `API` scope.

Table 5.1: Requirement Identifier Segments and Family Codes

Code	Meaning	Use
`REQ`	Requirement	Identifies the artifact as a requirement. Every requirement ID begins with this segment.

Requirement Identifier Segments and Family Codes (continued)

Code	Meaning	Use
`<scope>`	Project-defined scope	Optional uppercase qualifier for a governed subsystem, product, content type, or other area. Define every scope code before use; absence denotes project-wide scope.
`B`	Business	States the stakeholder outcome, mission, commercial purpose, policy goal, or business result the system shall enable.
`F`	Functional	States observable behavior, service, workflow, calculation, state change, or capability the system shall provide.
`Q`	Quality	States a measurable quality attribute or non-functional expectation, including accessibility, performance, reliability, resilience, usability, maintainability, or compatibility.
`I`	Interface	States a user, software, API, hardware, communication, protocol, or external-system boundary contract, including integration behavior at that boundary.
`S`	Security	States security, privacy, consent, authorization, auditing, secret-handling, or protected-data obligations maintained in a specialized security register.
`D`	Data	States data definition, authority, provenance, quality, integrity, ownership, lifecycle, retention, or source-of-truth obligations.
`C`	Constraint	States a binding implementation, platform, technology, format, storage, source-control, standards, compatibility, schedule, budget, or governance constraint.
`###`	Sequence	Three-digit, zero-padded stable sequence within the exact scope and family; assigned monotonically and never reused.

Classify a requirement by its primary obligation and verification concern. A requirement may support several concerns, but its identifier contains only one family code. Represent cross-cutting relationships through traceability rather than a compound family code.

An AI agent shall reuse an existing scope and family whenever their definitions fit. If no defined family accurately represents the requirement, the agent shall not silently invent or use a new code. It shall propose a unique uppercase code, full family name, decision boundary, rationale, and representative examples; check the complete requirements register and category catalog for semantic overlap and identifier collision; and obtain authorized-human approval. After approval, the agent shall add the family to Table 5.1, update the category catalog and AI guidance, and create or update the applicable register before assigning an identifier from it. The same approval and documentation rule applies to a new scope qualifier. Extending the convention shall not rename, renumber, or reuse an existing identifier.

5.3 Functional and Quality Requirements

Functional and quality requirements answer different questions and shall not be used interchangeably.

Central Distinction

A **functional requirement** states what behavior, service, calculation, state transition, or externally observable result the system shall provide. A **quality requirement** states how well that behavior or the system as a whole shall perform under stated conditions. Quality requirements require a measurable threshold, workload, environment, or other evaluation basis.

For example, "The system shall display the current request status" is functional. "The system shall display the current request status within two seconds under the reference workload" is a quality requirement. Avoid vague quality statements such as "the system shall be fast, secure, reliable, or easy to use." Replace them with measurable conditions and success criteria.

Quality requirements are also called quality-attribute or non-functional requirements. The term *quality requirement* is preferred here because these obligations describe behavior that shall be designed, implemented, and verified rather than an absence of function.

5.4 Requirement Status Model

Priority and status are independent. Priority records the requirement's importance or commitment; status records its progress through the delivery lifecycle. A status change shall not erase or replace the assigned priority.

Table 5.2: Requirement Status Definitions

Status	Meaning	In Version Rule
Open	The requirement is active, but implementation has not started. Deferred work remains Open until work begins or the requirement is retired.	Enter ∅.
In Progress	Implementation or integration work has started, but the requirement does not yet satisfy the definition of Done.	Enter ∅.
Done	The requirement is implemented; mapped tests pass; verification is complete; traceability is current; and the completed behavior is included in a controlled version.	Enter the first controlled version containing the completed requirement.

Status	Meaning	In Version Rule
Retired	The requirement is no longer active because it was superseded, made obsolete, rejected, or withdrawn. Retain its stable ID, original text, and traceability; record the rationale and any replacement in document history.	Retain the original implementation version if it was previously Done; otherwise enter ∅.

Only **Done** asserts completion. Do not remove a retired requirement or reuse its ID. A requirement may move from **Retired** only through an approved change that restores its active status and records the decision in document history.

5.5 Requirement Category Catalog

The following catalog distinguishes broad requirement families from their subtypes. In particular, performance, capacity, scalability, reliability, availability, resilience, recoverability, usability, accessibility, maintainability, supportability, testability, portability, compatibility, interoperability, security, privacy, safety, sustainability, and observability are quality attributes. A requirement concerning one of these attributes is a quality requirement when it specifies how well the system shall operate under stated conditions.

A project may organize a specialized quality attribute, such as security, privacy, or safety, in a separate register when it has distinct owners, review criteria, evidence, or assurance obligations. The separate register is a management view; it does not change the requirement's classification as a quality requirement. Likewise, interface, data, operational, and other families may contain both functional obligations and associated quality obligations. Use the primary purpose of each requirement to select its register and use links to represent cross-cutting concerns.

The catalog is intentionally broad. A project may use a family as a top-level register, create registers for selected subtypes, combine closely related groups, or omit groups that do not apply. Category choice shall not weaken the traceability or verification rules for individual requirements.

5.6 Domain Profile Impact on Requirements

Every Applicable domain profile shall be evaluated when developing or changing requirements. Each applicable profile concern shall result in one or more project requirements, a reference to an existing requirement, or a documented Not Applicable decision with rationale approved by an authorized human. Record the profile applicability decision in the Development Plan and retain project-specific obligations only in the applicable requirement registers. The profile itself is a tailoring guide, not a duplicate requirements specification. See Section 1.8 and the applicable profile appendix.

Table 5.3: Requirement Families, Subtypes, and Quality Attributes

Requirement Family	**Subtype or Quality Attribute**	**Typical Scope**
Business	Outcomes and policy	Organizational outcomes, value, policy goals, measurable benefits, cost targets, and mission objectives.
Stakeholder and user	Stakeholder need	Capabilities, outcomes, and constraints needed by a stakeholder group or external party.
Stakeholder and user	User need	User goals and externally visible outcomes expressed independently of a particular implementation.
Functional	System behavior	Services, calculations, decisions, state changes, workflows, and observable system responses.
Quality	General quality	Measurable characteristics that state how well a function or the system shall operate under stated conditions.
Quality	Performance	Response time, latency, throughput, processing time, and resource consumption.
Quality	Capacity	Supported data volume, transaction volume, concurrent users, devices, or retained records.
Quality	Scalability	Behavior as workload, data, users, or deployed instances increase or decrease.
Quality	Reliability	Probability or frequency of failure-free operation under stated conditions.
Quality	Availability	Required service availability, planned downtime, and permitted outage duration.
Quality	Resilience and fault tolerance	Continued or degraded operation when components, dependencies, or communication paths fail.
Quality	Recoverability and continuity	Recovery time, recovery point, restoration, backup, failover, and disaster-continuity objectives.
Quality	Usability	Task effectiveness, efficiency, learnability, error prevention, and user satisfaction.
Quality	Accessibility	Conformance and measurable support for users with visual, auditory, motor, speech, or cognitive disabilities.
Quality	Maintainability	Time, effort, modularity, analyzability, and change characteristics required for system modification.
Quality	Supportability	Diagnostic access, service procedures, field replacement, support hours, and support-team capabilities.

Continued on next page

Requirement Families, Subtypes, and Quality Attributes (continued)

Requirement Family	Subtype or Quality Attribute	Typical Scope
Quality	Testability	Controllability, observability, repeatability, test interfaces, and evidence needed to verify behavior.
Quality	Portability	Ability to install, move, or operate the system across specified environments or platforms.
Quality	Compatibility	Coexistence and correct operation with specified versions, devices, browsers, operating systems, or products.
Quality	Interoperability	Coordinated behavior, protocols, semantics, orchestration, and exchanges with external systems.
Quality	Security	Authentication, authorization, confidentiality, integrity, cryptography, auditing, and threat resistance.
Quality	Privacy	Collection, consent, purpose, minimization, access, disclosure, retention, deletion, and rights for personal data.
Quality	Safety	Hazard prevention, risk reduction, safe state, interlocks, alarms, and unacceptable-loss controls.
Quality	Sustainability	Energy use, resource efficiency, carbon reporting, hardware life, repair, reuse, and disposal objectives.
Quality	Observability	Logs, metrics, traces, health signals, diagnostic context, and measurable diagnostic coverage.
Interface	User interface	Human-system interaction, presentation, navigation, input, feedback, and interaction contracts.
Interface	Software and API	Operations, messages, schemas, errors, protocols, sequencing, and software boundary contracts.
Interface	Hardware and communication	Connectors, signals, electrical characteristics, device protocols, networks, and physical boundaries.
Data and information	Data definition and quality	Data definitions, schemas, provenance, accuracy, completeness, integrity, ownership, and permitted transformations.
Data and information	Lifecycle and governance	Creation, access, retention, archival, legal hold, disposition, deletion, and proof of destruction.
Operational	Operation and administration	Startup, shutdown, scheduling, configuration, provisioning, monitoring, backup, recovery, and incident response.

Continued on next page

Requirement Families, Subtypes, and Quality Attributes (continued)

Requirement Family	Subtype or Quality Attribute	Typical Scope
Operational	Deployment and platform	Approved platforms, topology, packaging, installation, configuration, upgrade, and runtime environment.
Operational	Infrastructure and environment	Network, compute, storage, facility, power, temperature, physical location, and environmental limits.
Transition	Migration and rollout	Data conversion, coexistence, rollout, cutover, rollback, legacy integration, and decommissioning.
Compliance and legal	Regulatory and standards	Obligations imposed by regulations, standards, certifications, policies, or assurance frameworks.
Compliance and legal	Legal and contractual	Licenses, warranties, intellectual-property terms, service agreements, procurement terms, and contractual duties.
Compliance and legal	Reporting and audit	Reports, evidence, audit trails, review access, content, timing, and record accountability.
Product support	Documentation	Required user, operator, administrator, developer, API, maintenance, and compliance documentation.
Product support	Training	Required curricula, exercises, qualifications, delivery modes, and demonstrated learner outcomes.
Commercial	Licensing and entitlement	Product editions, entitlement, metering, third-party components, subscriptions, and licensing restrictions.
AI and machine learning	Model and data controls	Model performance, data provenance, bias evaluation, explainability, human oversight, drift, and retraining controls.
Physical and hardware	Physical characteristics	Dimensions, mass, materials, connectors, tolerances, environmental ratings, and hardware behavior.
Constraint	Solution constraint	Mandated technologies, designs, budgets, schedules, standards, dependencies, or solution boundaries.

5.7 Creating a Project-Specific Category

Create an additional category when it gives requirements a distinct meaning, owner, review method, or assurance process. Define the category's scope and a unique family code using Section 5.2; duplicate one of the category-specific registers below; keep the standard columns; add the category to the category catalog; and update use cases, tests, traceability, verification, validation, document history, and the index as applicable. Document whether

the new category is top-level or a subtype of an existing category.

Do not create a category solely to avoid resolving an ambiguous requirement. If one statement contains multiple independently verifiable obligations, split it into singular requirements and assign each one a primary category.

5.8 Review Rules

Reject a requirement row if its priority, status, use-case mapping, or test-case mapping is empty. The **Status** field shall contain Open, In Progress, Done, or Retired. The **In Version** field shall follow the status definitions. Confirm that reciprocal mappings appear in the user-story, use-case, test, traceability, and verification registers.

5.9 Category-Specific Requirement Registers

The seven registers below are the template's default working set. They are selected for practical review and ownership, not as seven mutually exclusive top-level families. For example, the Security Requirements Register is a specialized quality-requirement register. Add, rename, combine, or remove registers to match the approved project taxonomy. Every retained register shall use the same traceability columns.

5.9.1 Business Requirements

Table 5.4: Business Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-B-001	The system shall enable an authorized analyst to prepare approved request-history data for analysis within five minutes for the reference data set.	Must	Open	∅	UC-003	TC-009

5.9.2 Functional Requirements

Table 5.5: Functional Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-F-001	The system shall accept a valid service request and assign it a unique identifier.	Must	Done	1.0	UC-001	TC-001
REQ-F-002	The system shall display the current status and most recent update for a submitted request.	Should	Done	1.1	UC-002	TC-002
REQ-F-003	The system shall export request history in the approved interchange format.	Would	Open	∅	UC-003	TC-004

5.9.3 Quality Requirements

Table 5.6: Quality Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-Q-001	The system shall display a request status within two seconds under the reference workload.	Could	Done	1.2	UC-002	TC-003

© Computer Club

5.9.4 Interface Requirements

Table 5.7: Interface Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-I-001	The system shall exchange request-history exports using the approved, versioned interface schema.	Should	In Progress	∅	UC-003	TC-005

5.9.5 Security Requirements

Table 5.8: Security Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-S-001	The system shall permit request-history exports only to authenticated actors assigned the approved export role.	Must	Done	1.0	UC-003	TC-006

5.9.6 Data Requirements

Table 5.9: Data Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-D-001	The system shall retain each request status transition with its timestamp, actor identifier, previous status, and new status.	Must	Done	1.1	UC-002, UC-003	TC-007

5.9.7 Constraint Requirements

Table 5.10: Constraint Requirements Register

ID	Requirement	Priority	Status	In Version	Use Case(s)	Test Case(s)
REQ-C-001	The system shall support the deployment platform and browser versions identified in the approved deployment baseline.	Must	Done	1.0	UC-001, UC-002, UC-003	TC-008

Chapter 6

Use Cases

6.1 How to Use This Chapter

Describe goal-oriented interactions at the system boundary. Give every use case a stable ID, its own table, and at least one reciprocal requirement mapping. Include preconditions, trigger, main flow, alternate or exception flows, and postconditions. Maintain the canonical relationships among stories, requirements, use cases, and tests in Chapter 15.

6.2 High-Level Use Case Diagram

Show the system boundary, primary actors, and major user goals before the detailed use-case tables. Replace the generic example with project-specific actors and use cases while preserving the stable use-case identifiers. Figure 6.1 provides the template example.

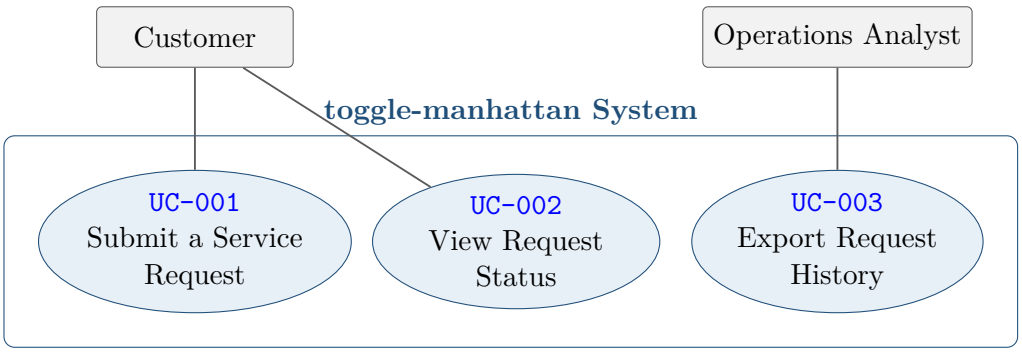


Figure 6.1: Generic high-level use case diagram.

6.3 Submit a Service Request

Identifier	UC-001

Mapped requirements	REQ-F-001, REQ-C-001
Goal	Submit a valid request and receive a unique identifier.
Primary actor	Customer
Preconditions	The customer can access the request form.
Trigger	The customer chooses to submit a new request.
Main flow	The customer enters required data and submits it; the system validates the data, creates the request, and displays its identifier.
Alternate flow	If required data is invalid, the system identifies the affected fields and creates no request.
Postconditions	A valid request exists with a unique identifier and an initial status.

6.4 View Request Status

Identifier	UC-002
Mapped requirements	REQ-F-002, REQ-Q-001, REQ-D-001, REQ-C-001
Goal	Review the current state of a submitted request.
Primary actor	Customer
Preconditions	A request with a known identifier exists.
Trigger	The customer opens or refreshes the request status.
Main flow	The system retrieves and displays the request status and most recent update.
Alternate flow	If the identifier is unknown, the system reports that no matching request was found.
Postconditions	The request remains unchanged; its current status has been presented.

6.5 Export Request History

Identifier	UC-003
Mapped requirements	REQ-B-001, REQ-F-003, REQ-I-001, REQ-S-001, REQ-D-001, REQ-C-001
Goal	Export request history for authorized analysis.
Primary actor	Operations analyst
Preconditions	The actor is authorized and matching requests exist.
Trigger	The actor selects an export range and format.

Main flow	The system validates authorization, generates the export, and makes the result available to the actor.
Alternate flow	If no records match, the system produces an empty result with an explanatory status.
Postconditions	An export artifact exists; source request records are unchanged.

Chapter 7

Architecture Overview

7.1 How to Use This Chapter

Explain the forces that shape the design, the major decisions, alternatives considered, and consequences. Use the 4+1 views only when they help readers answer different architectural questions.

7.2 Decision Template and Example

ARC-001: *[Decision]*; **Drivers:** *[quality attributes and constraints]*; **Rationale:** *[why selected]*; **Consequences:** *[benefits, trade-offs, and risks].* Example: use an asynchronous message queue to protect request processing from transient downstream outages; this improves resilience but introduces eventual consistency.

Chapter 8

User Interface Design

8.1 How to Use This Chapter

Describe personas, primary screens or workflows, interaction rules, feedback, accessibility, and essential states. Include wireframes or prototypes where they resolve meaningful ambiguity.

8.2 Example

Persona: a busy operations coordinator needs to submit and track requests quickly. The request screen groups required fields, identifies inline validation errors, confirms successful submission, supports keyboard operation, and exposes empty, loading, error, and completed states.

Chapter 9

Logical View

9.1 How to Use This Chapter

Model the domain concepts, responsibilities, relationships, data ownership, and invariants without committing to code-level structure. Use a class, entity, or conceptual diagram where useful.

9.2 Example

Request has an identifier, submitter, status, and submitted time. A request has one or more *Request Items*. Invariants: every identifier is unique; every request has a valid status; an approved request cannot return to Draft without an authorized transition.

Chapter 10

Process View

10.1 How to Use This Chapter

Describe runtime behavior, concurrency, state transitions, workflows, error handling, and significant sequences. Use activity, sequence, or state diagrams when prose is insufficient.

10.2 Example State Model

Request states are Draft, Submitted, Under Review, Approved, Rejected, and Cancelled. Submission moves Draft to Submitted only when required data is valid. Approval and rejection are terminal except for an authorized correction workflow.

Chapter 11

Development View

11.1 How to Use This Chapter

Show how source code, modules, services, libraries, and build artifacts are organized. State dependency rules so that implementers do not create forbidden coupling.

11.2 Example

Organize a service into presentation/API, application workflow, domain model, and infrastructure adapters. The API calls application services; application services depend on domain abstractions; infrastructure implements those abstractions. Domain logic must not depend on the web framework or database driver.

Chapter 12

Physical View

12.1 How to Use This Chapter

Describe deployment nodes, networks, environments, runtime dependencies, trust boundaries, data stores, and operations constraints. Include a deployment diagram for distributed systems.

12.2 Example

A browser communicates over TLS with an API service. The API service uses a private database network; backups are encrypted and access is limited to the operations role. Define availability targets, monitoring, and recovery responsibilities.

Chapter 13

Implementation

13.1 How to Use This Chapter

Document the selected technology baseline, build and run instructions, coding conventions that enforce architecture, configuration, and definition of done. Keep design rationale in Architecture Overview.

13.2 Example

The implementation uses [language], [framework], and [database]. Build with `[command]`; run with `[command]`. Configuration comes from [approved source]; credentials are never stored in source control. Completion requires passing tests, reviewed traceability, and approved evidence.

Chapter 14

Tests

14.1 How to Use This Chapter

Define repeatable test cases, levels, environments, data, entry conditions, procedures, expected results, and result recording. Tests produce evidence; they do not replace verification or validation.

14.2 Example Test Cases

Table 14.1: Test Case Examples

ID	Mapped Requirement	Procedure	Expected Result
TC-001	REQ-F-001	Submit a request containing valid required data.	The system creates one request and displays a unique identifier.
TC-002	REQ-F-002	Open the status for a known request.	The current status and most recent update are displayed.
TC-003	REQ-Q-001	Measure 20 status requests under the reference workload.	Every measured response is displayed within two seconds.
TC-004	REQ-F-003	Export a known request set in the approved format.	The export contains the expected history without changing source records.
TC-005	REQ-I-001	Export representative request history and validate the result against the approved interface schema version.	The export conforms to the approved schema and declares the expected schema version.

ID	Mapped Requirement	Procedure	Expected Result
TC-006	REQ-S-001	Attempt the same export as an authorized actor, an authenticated actor without the export role, and an unauthenticated actor.	Only the authorized actor receives an export; both unauthorized attempts are denied and recorded.
TC-007	REQ-D-001	Perform two status transitions and inspect the stored history and exported record.	Each transition retains the expected timestamp, actor identifier, previous status, and new status.
TC-008	REQ-C-001	Execute the critical use-case test suite on every platform and browser version in the approved deployment baseline.	Every critical use case passes on every approved platform and browser version.
TC-009	REQ-B-001	Time an authorized analyst preparing an approved export from the reference data set without manual reconstruction.	The analyst obtains analysis-ready request-history data within five minutes.

Chapter 15

Traceability

15.1 How to Use This Chapter

Maintain the canonical mapping from stakeholder need or approved constraint through requirement, design decision, implementation, test, verification, validation, and release evidence. Review both missing backward links and missing forward links.

15.2 Traceability Register

Requirement and use-case chapters shall contain reciprocal mappings. An unlinked requirement may be gold-plating; an untested requirement or unmapped use case is incomplete.

Table 15.1: Traceability Register

User Story	Requirement	Use Case	Test Case	Verification	Validation
US-001	REQ-F-001	UC-001	TC-001	VER-001	VAL-001
US-002	REQ-F-002	UC-002	TC-002	VER-002	VAL-002
US-002	REQ-Q-001	UC-002	TC-003	VER-003	VAL-002
US-003	REQ-F-003	UC-003	TC-004	VER-004	VAL-003
US-003	REQ-I-001	UC-003	TC-005	VER-005	VAL-003
US-003	REQ-S-001	UC-003	TC-006	VER-006	VAL-003
US-002	REQ-D-001	UC-002	TC-007	VER-007	VAL-002
US-003	REQ-D-001	UC-003	TC-007	VER-007	VAL-003
US-001	REQ-C-001	UC-001	TC-008	VER-008	VAL-001
US-002	REQ-C-001	UC-002	TC-008	VER-008	VAL-002
US-003	REQ-C-001	UC-003	TC-008	VER-008	VAL-003
US-003	REQ-B-001	UC-003	TC-009	VER-009	VAL-003

Chapter 16

Verification

16.1 How to Use This Chapter

Show that each requirement conforms to its specification using test, inspection, analysis, or demonstration. Record the method, evidence reference, responsible reviewer, result, and any waiver. Verification answers "Did we build the system right?"

16.2 Verification Register

Table 16.1: Verification Register

ID	Requirement	Method	Required Evidence	Status
VER-001	REQ-F-001	Test	Passing TC-001 execution record showing creation of one request with a unique identifier.	**PASS**
VER-002	REQ-F-002	Test	Passing TC-002 execution record showing the current status and most recent update.	**PASS**
VER-003	REQ-Q-001	Analysis and test	Timing analysis and passing TC-003 measurements under the reference workload.	**PASS**
VER-004	REQ-F-003	Inspection and test	Export inspection and passing TC-004 execution record.	**PENDING**
VER-005	REQ-I-001	Inspection and test	Schema-validation report and passing TC-005 execution record.	**PENDING**
VER-006	REQ-S-001	Analysis and test	Access-control analysis, denial records, and passing TC-006 execution record.	**PASS**
VER-007	REQ-D-001	Inspection and test	Stored-history inspection and passing TC-007 execution record.	**PASS**

Continued on next page

Verification Register (continued)

ID	Requirement	Method	Required Evidence	Status
VER-008	REQ-C-001	Test	Approved deployment baseline and passing TC-008 compatibility results.	**PASS**
VER-009	REQ-B-001	Test	Timed analyst demonstration and passing TC-009 execution record.	**PENDING**

Chapter 17

Validation

17.1 How to Use This Chapter

Evaluate whether the completed system solves stakeholder needs in its intended context. Define realistic scenarios, participant roles, success criteria, observations, and disposition of requested changes. Validation answers "Did we build the right system?"

17.2 Validation Register

Table 17.1: Validation Register

ID	Related Story	Scenario	Success Criterion	Status
VAL-001	US-001	A customer submits a realistic service request without coaching.	The customer receives a unique identifier and confirms that the workflow supports the intended task.	PENDING
VAL-002	US-002	A customer checks the progress of an existing request.	The customer correctly identifies the current status and latest update and judges the response timely.	PENDING

Continued on next page

Validation Register (continued)

ID	Related Story	Scenario	Success Criterion	Status
VAL-003	US-003	An authorized analyst exports a representative request history.	Within five minutes, the analyst opens the approved format, finds the expected records, and confirms source records were unchanged.	PENDING

Chapter 18

Release

18.1 How to Use This Chapter

Identify the release and define contents, approvals, evidence, known limitations, deployment instructions, rollback or recovery approach, and support ownership.

18.2 Example Checklist

Confirm the version and artifact IDs are current; all required traceability is complete; required tests, verification, and validation pass or have approved waivers; release notes and known limitations are published; and the release authority approves the exact package.

Bibliography

[1] *ISO/IEC/IEEE 15288:2023, Systems and Software Engineering—System Life Cycle Processes*, International Organization for Standardization Std. ISO/IEC/IEEE 15 288:2023, 2023. [Online]. Available: https://www.iso.org/standard/81702.html

[2] *ISO/IEC/IEEE 12207:2026, Systems and Software Engineering—Software Life Cycle Processes*, International Organization for Standardization Std. ISO/IEC/IEEE 12 207:2026, 2026. [Online]. Available: https://www.iso.org/standard/90219.html

Appendix A

Document Authoring Guide

A.1 How to Use This Appendix

Use this appendix to maintain authoring conventions, repository locations, diagram sources, review rules, tool installation, and PDF build instructions. Update it whenever the document structure or toolchain changes.

A.2 Template Provenance

This template was developed by **Darian Muresan, Ph.D.**, in conjunction with the AI agents **OpenAI Codex** and **Anthropic Claude Code**. Questions, corrections, and proposed improvements may be sent to dmuresan@stevens.edu, darian@dmmd.net, or darian@pictomic.com.

Keep the template attribution distinct from the project-specific document owner and organization. A derived project may replace the project metadata while retaining this provenance statement for the underlying template.

A.3 AI-Assisted Authoring

The authoritative instructions for AI-assisted architecture work, including task routing, limited context loading, workflows, example prompts, completion checks, and human-review boundaries, are contained in Chapter C. This appendix retains installation, build, file, and LaTeX authoring guidance that applies to both human authors and AI agents.

A.4 File Naming

Every project-owned file created for this document shall begin with the project file namespace, followed by a Pascal-style responsibility name. The namespace is defined once in `tgmManual.tex`. The defining macro is `FileNamespace`, and this template sets it to `tgm`. Example chapter files include `tgmRequirements.tex`, `tgmTests.tex`, and `tgmValidation.tex`.

The convention applies to chapter files, diagram sources, generated project figures, evidence files, and other project-owned artifacts. A tool-mandated or externally maintained file name may remain an explicitly documented exception, but shall not establish a second project naming convention.

Before creating or renaming files for a derived project, an AI agent shall ask an authorized human for the project file namespace or confirm a namespace that is already explicitly configured. The agent shall not invent one. After it is confirmed, replace `\FileNamespace` and apply the namespace consistently to every new project-owned file.

A.4.1 Document-Authoring Requirements

The following requirements govern the template itself and every project derived from it. They are verified by repository inspection and by compiling the document.

Table A.1: Document-Authoring Requirements

ID	Requirement	Verification
DOC-FILE-001	Except for an explicitly documented tool-mandated discovery file, every new project-owned file shall be named `<nameSpace><FileName>.<extension>`, using the human-approved project namespace and a descriptive Pascal-style file name.	Inspect all newly created project files.
DOC-TIKZ-001	Every TikZ source shall be stored under `TikZ/<includingFileBaseName>/` and shall be named `<nameSpace><DrawingName>.tex`. The directory name shall exactly match the base name of the LaTeX file that contains its `input` command.	Compare each TikZ path with its including LaTeX file and compile the document.
DOC-AI-001	Before creating or renaming any project-owned file, an AI agent shall ask for or confirm the project `<nameSpace>` and shall apply the confirmed namespace to all files it creates.	Review the AI interaction record and inspect created file names.
DOC-AI-002	Every repository-level AI instruction file shall point to `tgmAIAgentGuide.tex` and the relevant authoritative LaTeX sources, shall instruct the agent to read them before architecture-related work, and shall not duplicate their rules. Tool-mandated discovery file names are documented exceptions to DOC-FILE-001.	Inspect each repository AI instruction file and compare it with the authoritative LaTeX guidance.

A.5 One Responsibility per Chapter

A chapter answers one major engineering question. When content belongs to another responsibility, link to its authoritative chapter instead of copying it. In particular, tests define procedures, verification records requirement compliance, validation evaluates stakeholder use, and release decides whether the evidence is sufficient.

A.6 Adding a Chapter

Create the namespaced file, begin it with a chapter title and stable chapter label, add useful index terms, and include it once from `tgmManual.tex`. Add or change artifact mappings only in the canonical traceability matrix.

```
\chapter{Example Chapter}
\ChapterLabel{ExampleChapter}
\index{example}
\index{Chapter!Example Chapter}

Chapter content goes here.
```

A.7 Diagram Authoring with TikZ

TikZ is the diagram-authoring tool for this template. Authors shall use TikZ for architecture diagrams because the editable source remains in LaTeX, text uses the document fonts, output stays vector based, and diagram changes can be reviewed with the surrounding specification. The main document loads TikZ and a common set of libraries for arrows, positioning, geometric shapes, fitting, and coordinate calculations.

Every TikZ source shall use the following directory hierarchy:

`TikZ/<includingFileBaseName>/<nameSpace><DrawingName>.tex`

The directory name shall exactly match the base name of the LaTeX file that contains the diagram's `\input` command. The TikZ file name shall begin with the configured project namespace and describe the drawing. For example, the Introduction chapter includes its V-model from:

`TikZ/tgmIntroduction/tgmVModel.tex`

Create additional including-file directories as required; do not place TikZ sources beside chapter files or in generated-image directories.

The TikZ source file should contain the `tikzpicture` environment. The owning chapter shall include that source in a numbered figure, provide a descriptive caption, place a stable label after the caption, and reference the figure from the chapter text:

```
As shown in Figure~\ref{Figure:HighLevelUseCase}, ...

\begin{figure}[htbp]
  \centering
  \input{TikZ/tgmUseCases/tgmUseCaseDiagram.tex}
  \caption{High-level use-case diagram.}
  \label{Figure:HighLevelUseCase}
\end{figure}
```

A.8 Building the PDF

TikZ diagrams compile directly when their source files are included by the owning chapters. The toolchain requires:

- a LaTeX distribution containing TikZ/PGF, `pdflatex`, `bibtex`, and `latexmk`;
- the `makeglossaries` and `makeindex` commands.

A.8.1 Installing the Tools on macOS

The full MacTeX distribution is recommended because it includes the LaTeX tools and packages used by this template. Complete these steps from an administrator account:

1. Open Terminal and install Apple's command-line tools, which provide `make`:

   ```
   xcode-select --install
   ```

   Complete the installer dialog before continuing.

2. Install Homebrew by following the current instructions at https://brew.sh/. Then install MacTeX, which includes TikZ:

   ```
   brew install --cask mactex
   ```

   MacTeX may instead be downloaded directly from https://tug.org/mactex/.

3. Close and reopen Terminal so the MacTeX command-line tools are added to the system path.

4. Verify that every required command is available:

   ```
   command -v make pdflatex bibtex \
     makeglossaries makeindex latexmk
   kpsewhich tikz.sty
   ```

   Each command shall print a path. A missing line identifies the tool or TikZ package that still needs to be installed or added to the system path.

A.8.2 Installing the Tools on Windows

Use a current Windows 10 or Windows 11 computer and complete these steps:

1. Download and run the Basic MiKTeX Installer from https://miktex.org/howto/install-miktex. Choose a private per-user installation when appropriate and select *Always install missing packages on-the-fly* for a computer with Internet access.

2. Open MiKTeX Console, check for updates, and install them. If `latexmk` is not found during verification, use the Packages page in MiKTeX Console to install the `latexmk` package.

3. Close and reopen PowerShell, then verify the required commands:

   ```
   Get-Command pdflatex,bibtex
   Get-Command makeglossaries,makeindex,latexmk
   kpsewhich tikz.sty
   ```

   Each tool shall produce a command record. A command-not-found error identifies a missing installation or system-path entry.

On an organization-managed computer, authors shall ask the system administrator before installing software or changing a managed execution policy.

A.8.3 Installing the Tools on Ubuntu or Debian Linux

Open a terminal and install TikZ/PGF, GNU Make, Latexmk, and a TeX Live set that contains the other packages used by this template:

```
sudo apt update
sudo apt install make latexmk texlive-pictures \
  texlive-latex-extra \
  texlive-fonts-recommended texlive-bibtex-extra \
  texlive-extra-utils
```

Verify the required commands:

```
command -v make pdflatex bibtex \
  makeglossaries makeindex latexmk
kpsewhich tikz.sty
```

For another Linux distribution, install the equivalent TikZ/PGF, GNU Make, Latexmk, and full or extended TeX Live packages with its package manager.

A.8.4 AI-Agent Installation

An AI coding agent can inspect the host, identify missing commands, request approval for system changes, install the appropriate packages, and verify the build. Give the agent the project directory and the following reusable request:

```
In this LaTeX project, inspect the build files and determine
which operating system and package manager are available.
Verify TikZ with ‘kpsewhich tikz.sty‘, and verify make,
pdflatex, bibtex, makeglossaries, makeindex, and latexmk.
Install only missing tools using the
platform’s standard package manager. Ask for approval before
administrator-level, system-wide, or networked installation.
Do not replace project sources or remove user files. After
installation, compile included TikZ sources, run the full platform
build, inspect the log for errors, unresolved
references, and overfull boxes, and report the generated PDF.
```

If the agent cannot install software, ask it to report the exact missing command, the attempted verification, and platform-specific installation instructions. Never give an agent an unrestricted instruction to delete build directories or modify system security policy.

A.8.5 Building on macOS and Linux

From the template directory, run:

```
make -f makefileMacLinux pdf
```

Use `make -f makefileMacLinux clean` to remove generated build files while preserving authored sources and the final PDF.

A.8.6 Building with Windows PowerShell

From PowerShell in the template directory, run:

```
.\makefilePowerShell.ps1 -Action pdf
```

Use `-Action clean` in place of `-Action pdf` to remove generated build files. If local execution policy prevents scripts from running, use an approved organizational policy or allow scripts only for the current PowerShell process:

```
Set-ExecutionPolicy -Scope Process Bypass
.\makefilePowerShell.ps1 -Action pdf
```

A.8.7 Manual Build Sequence

When the platform build script is unavailable, run the following commands from the template directory. The final two LaTeX passes resolve tables of contents, citations, glossary links, the index, and cross-references.

Included TikZ sources require no separate rendering command; `pdflatex` compiles them with their owning chapters:

```
pdflatex -interaction=nonstopmode -halt-on-error tgmManual.tex
bibtex tgmManual
makeglossaries tgmManual
makeindex tgmManual.idx
pdflatex -interaction=nonstopmode -halt-on-error tgmManual.tex
pdflatex -interaction=nonstopmode -halt-on-error tgmManual.tex
```

A.8.8 Build Verification and Troubleshooting

Confirm that the output PDF opens, all expected chapters and appendices appear, the glossary and index are populated, hyperlinks reach their targets, and no table or figure is clipped. Search `tgmManual.log` for fatal errors, undefined references, missing files, and overfull boxes. Common remedies are:

- open a new terminal after installation so its search path is refreshed;
- on macOS, verify that `/Library/TeX/texbin` is on the search path;
- rerun the complete build after adding citations, glossary entries, index entries, figures, labels, or references;
- compile and inspect the owning chapter after changing a TikZ source;
- install the named LaTeX package when the log reports a missing `.sty` file.

A.9 Authoring Rules

- Add the newest approved change to the top of the document history.
- Introduce specialized terms through the glossary and use them consistently.
- Give every controlled artifact a stable explicit ID.
- Use `shall` only for binding requirements.
- Keep normative requirements separate from explanation and examples.
- Apply the approved project namespace to every new project-owned file as specified in the document-authoring requirements.

- Use TikZ for all diagrams. Store every source in `TikZ/<includingFileBaseName>/` and name it `<nameSpace><DrawingName>.tex`.
- Include each TikZ source from its owning chapter in a numbered figure with a caption and stable label, and reference that figure in the chapter text.
- Compile and inspect the PDF after structural or table changes.

A.10 Citations and Bibliography

Store controlled reference records in `tgmReferences.bib`. Cite a source from a chapter with a stable BibTeX key:

```
\cite{ISO29148}
```

The main file applies the style and prints one document-level bibliography:

```
\bibliographystyle{IEEEtran}
\bibliography{tgmReferences}
```

Both platform build scripts run BibTeX automatically.

Bibliography entries support claims and definitions; they do not replace project-owned requirements, architecture decisions, test procedures, or approval evidence.

A.11 Selective Builds

For a large project, LaTeX's `\includeonly` command may select chapters during drafting. The main file contains a commented example:

```
% \includeonly{
%   prologue,
%   tgmRequirements,
%   tgmTests
% }
```

Remove the leading comment characters and list include names without the `.tex` extension. LaTeX can retain cross-references and citations from omitted chapters by reading their existing auxiliary files, but those files may be absent or stale. Selective builds are therefore drafting aids only. Hyperref may also report missing destinations for links into deliberately omitted chapters; these draft warnings shall disappear in the full build. Comment out the complete `\includeonly` block and run a full build before verification or release so missing chapters, references, glossary entries, citations, and evidence cannot be hidden.

Appendix B

Embedded Systems Tailoring Profile

B.1 How to Use This Appendix

This appendix tailors the generic architecture template for systems in which software interacts directly with dedicated hardware, constrained computing resources, physical processes, or real-time events. It applies to bare-metal firmware, real-time operating system (RTOS) products, embedded Linux systems, programmable logic, hardware–software products, and other cyber-physical systems.

The profile augments the core template; it does not create a second architecture specification. Project-specific requirements, interfaces, decisions, diagrams, tests, and evidence shall remain in their authoritative core chapters. When this profile applies, retain this appendix as the authoritative tailoring guide, record the applicability decision in Chapter 2, and use the additional concerns below when creating and reviewing the core content. Remove this appendix from an instantiated project only when an authorized human records that the profile is not applicable.

B.2 Applicability

Apply this profile when one or more of the following conditions is present:

- software executes on a dedicated processor, microcontroller, system on chip, field-programmable device, or purpose-built computing module;
- behavior depends on interrupts, deadlines, deterministic scheduling, physical signals, sensors, actuators, or control loops;
- CPU, memory, storage, bus bandwidth, energy, or thermal resources are materially constrained;
- the product requires hardware bring-up, boot firmware, board support, device drivers, manufacturing provisioning, calibration, or field firmware updates; or

- safety, reliability, environmental, physical-security, certification, or component-lifecycle concerns affect the architecture.

The applicability decision shall identify the embedded platform class, the reason the profile applies, the hardware revisions in scope, and any other domain profiles used concurrently. A project may combine this profile with a future safety-critical, high-security, cloud, mobile, or machine-learning profile. Each profile remains guidance; controlled project facts stay in the core chapters.

B.3 Core-Chapter Tailoring Map

Use Table B.1 to place each embedded concern in one authoritative location.

Table B.1: Embedded-Systems Core-Chapter Tailoring Map

Core Chapter	**Embedded-System Additions**
Development Plan	Record the profile decision, hardware and firmware roles, toolchain ownership, hardware availability, prototypes, certification gates, manufacturing dependencies, and component-lifecycle risks.
System Context	Define the hardware–software boundary, physical environment, sensors, actuators, operators, service tools, external buses, power sources, and upstream or downstream systems.
User Stories and Use Cases	Include installation, startup, shutdown, calibration, diagnostics, maintenance, update, recovery, degraded operation, manufacturing, and service interactions where applicable.
Requirements	Add timing, resource, power, interface, boot, recovery, safety, security, reliability, environmental, diagnostics, update, manufacturing, and regulatory obligations.
Architecture Views	Document hardware–software allocation, firmware layers, execution contexts, states, timing, memory, deployment, data and signal flow, power modes, faults, and recovery.
Implementation	Identify processor and board revisions, boot chain, BSP, HAL, RTOS or operating system, compiler, linker, SDK, libraries, generated code, linker scripts, build variants, image formats, and programming tools.
Tests, Verification, and Validation	Add target execution, static analysis, timing analysis, SIL, PIL, HIL, fault injection, power, environmental, hardware-revision, update, recovery, and retained-evidence activities.
Release	Identify released hardware revisions, firmware images, hashes, configuration data, programming instructions, compatibility, rollback, manufacturing release, service provisions, and known limitations.

B.4 Hardware–Software Boundary and Allocation

The System Context and architecture chapters shall identify every processor, memory device, programmable component, peripheral, sensor, actuator, bus, power domain, and external physical interface that materially constrains the software. Define which behavior is implemented in hardware, firmware, programmable logic, operating-system services, application software, or an external system. Record alternatives and trade-offs for significant allocation decisions.

For each allocated function, identify its driving requirements, execution platform, inputs, outputs, timing needs, failure behavior, update boundary, and verification method. Do not treat a vendor module, BSP, HAL, RTOS, device driver, or generated component as an unexamined implementation detail when it affects required behavior.

B.5 Embedded Requirement Categories

Table B.2 is a checklist of common embedded requirement categories. It does not introduce new requirement ID families. Classify each requirement in the existing family that best describes its primary obligation. Propose a new family only through the controlled procedure in Chapter 5. Apply the domain-profile translation rule in Section 5.6: every applicable embedded concern shall map to a project requirement, an existing requirement, or an approved Not Applicable decision with rationale.

Table B.2: Embedded-System Requirement Categories

Category	**Questions the Requirements Shall Resolve**
Functional behavior	What observable control, sensing, computation, communication, diagnostic, maintenance, and recovery behavior is required?
Timing and real time	What deadlines, periods, latency, jitter, sequencing, startup, shutdown, and worst-case response limits apply?
Processing resources	What CPU utilization, execution-time, core-affinity, interrupt-latency, and scheduling limits apply?
Memory and storage	What flash, RAM, stack, heap, nonvolatile storage, retention, endurance, and memory-protection limits apply?
Power and thermal	What operating, sleep, peak, average, battery-life, wake-up, brownout, thermal, and power-transition behavior is required?
Hardware interfaces	What electrical levels, pin assignments, connectors, buses, protocols, rates, units, ranges, scaling, resolution, endianness, and error behavior apply?
Boot and initialization	What boot sources, ordering, self-tests, startup times, configuration checks, and failure responses are required?
Reset and recovery	What events cause reset, retry, failover, degraded mode, safe state, rollback, or operator intervention?
Safety	What hazards, safety functions, safe states, diagnostic coverage, independence, and integrity constraints apply?

Embedded-System Requirement Categories (continued)

Category	Questions the Requirements Shall Resolve
Security	What root of trust, secure boot, image authentication, device identity, key provisioning, access control, debug-port, anti-rollback, and physical-attack provisions apply?
Reliability and availability	What service life, duty cycle, fault tolerance, watchdog, data-integrity, availability, maintainability, and recovery targets apply?
Environmental	What temperature, humidity, vibration, shock, altitude, electromagnetic, contamination, ingress, and storage conditions apply?
Diagnostics and observability	What logs, counters, health indicators, fault codes, traces, test points, telemetry, and service interfaces are required?
Calibration and configuration	What values are configurable or calibrated; who controls them; how are range, integrity, compatibility, and provenance maintained?
Firmware update	What transport, authentication, atomicity, compatibility, interruption, rollback, recovery, audit, and field-service behavior applies?
Manufacturing and provisioning	What programming, serialization, key injection, calibration, end-of-line testing, traceability, and acceptance data is required?
Regulatory and certification	What standards, evidence, configuration controls, independence, analysis, and approval obligations apply?
Component lifecycle	What approved components, alternates, lead-time, obsolescence, silicon errata, vendor support, and redesign constraints apply?

Each embedded requirement shall retain the normal priority, lifecycle status, In Version value, use-case mapping, and test-case mapping. Quantitative limits shall state units, applicable operating conditions, tolerance, and verification method. Avoid terms such as “fast,” “low power,” or “robust” without a measurable acceptance boundary.

B.6 Required Architecture Views

Select views according to risk and complexity, and explain any omitted view that would normally apply. Create diagrams with TikZ and store them under the owning core chapter.

Table B.3: Candidate Embedded-System Architecture Views

View	Content
Hardware block view	Processing, memory, programmable logic, peripherals, clock, reset, power, sensors, actuators, and external connections.
Hardware–software allocation	Required functions allocated to hardware, firmware layers, programmable logic, or external systems.
Firmware layering	Bootloader, BSP, HAL, drivers, RTOS or kernel, middleware, services, application, and diagnostic boundaries.
Boot and update sequence	Trust chain, initialization order, self-tests, configuration loading, update, rollback, and recovery paths.

Candidate Embedded-System Architecture Views (continued)

View	Content
Execution-context view	Cores, processes, tasks, threads, interrupt service routines, priorities, scheduling, synchronization, and shared resources.
State and mode view	Operational, maintenance, calibration, degraded, safe, sleep, update, and fault states with permitted transitions.
Timing view	Periods, deadlines, latency, jitter, sequencing, bus transactions, control-loop timing, and worst-case paths.
Memory view	Address spaces, partitions, executable regions, stacks, heaps, DMA buffers, retained data, memory protection, and nonvolatile layout.
Data and signal flow	Units, scaling, sample rates, filtering, validity, ownership, persistence, and error propagation.
Deployment view	Firmware images and configuration deployed to processors, cores, boards, hardware revisions, and manufacturing or field tools.
Power and thermal view	Power domains, modes, transitions, wake sources, consumers, thermal limits, throttling, and shutdown behavior.
Fault-containment view	Fault-detection boundaries, watchdogs, redundancy, degraded operation, safe states, containment, and recovery.

B.7 Interfaces and Protocols

For every hardware or communication interface, document the physical and logical contract together. Include connector and pin identity, direction, electrical characteristics, protocol and version, topology, addressing, message or register definition, data type, units, range, resolution, scaling, endianness, update rate, timing, initialization, ownership, concurrency, timeouts, integrity checks, invalid values, error handling, compatibility, and verification method where applicable.

An interface table or referenced interface-control document shall have a controlled identifier and version. Each interface requirement and test shall link to that authority. Do not infer an electrical limit, register behavior, reserved bit, or protocol tolerance without controlled evidence.

B.8 Resource Budgets

Maintain resource budgets whenever architecture feasibility depends on a finite embedded resource. Table B.4 is a generic pattern; create one row for each controlled allocation and reconcile component allocations with the system total.

Table B.4: Generic Embedded Resource Budget

Resource	**Owner**	**Allocation**	**Measured or Estimated**	**Requirement / Evidence**
CPU time	[task or component]	[limit and units]	[value, method, conditions]	[requirement and test IDs]
Flash	[image or component]	[bytes or percent]	[map-file value]	[requirement and evidence]
RAM	[task or component]	[bytes or percent]	[static and peak value]	[requirement and evidence]
Stack	[task or context]	[bytes]	[worst observed or analyzed]	[requirement and evidence]
Bus bandwidth	[interface or flow]	[rate and utilization]	[value and conditions]	[requirement and test IDs]
Power or energy	[mode or component]	[power, current, or energy]	[value and conditions]	[requirement and test IDs]
Thermal margin	[component or mode]	[temperature or margin]	[value and conditions]	[requirement and evidence]

Every budget shall state the measurement or estimation conditions, margin policy, uncertainty, owner, update trigger, and response to an exceeded allocation. Budget evidence shall distinguish analysis, simulation, measurement, and worst-case guarantee.

B.9 Safety, Reliability, and Fault Management

Identify hazards, failure modes, fault sources, detection mechanisms, containment boundaries, watchdogs, redundancy, degraded modes, safe states, recovery behavior, diagnostic coverage, and residual risk where applicable. Trace every credited mechanism to requirements and verification evidence.

Do not claim a safety integrity level, diagnostic coverage, failure rate, independence, or risk acceptance without the applicable analysis and authorized approval. When a dedicated safety profile or regulated process applies, use it together with this embedded profile and record precedence for any conflicting rule.

B.10 Embedded Security

Define the trust boundary and lifecycle from manufacturing through decommissioning. Address device identity, root of trust, secure boot, image authentication, signing and key custody, provisioning, credential storage, debug and test access, privileged interfaces, update

authorization, anti-rollback, protected configuration, security logging, recovery, physical attack assumptions, vulnerability response, and secure decommissioning as applicable.

Security controls shall be mapped to threats, requirements, tests, retained evidence, and operational responsibilities. Secrets and production keys shall not be placed in this architecture document, source repository, AI prompt, or test artifact.

B.11 Toolchain and Configuration Control

Record the controlled hardware revisions, processor variants, compiler and linker versions, optimization and warning settings, linker scripts, BSP, HAL, RTOS or operating-system version, SDK, libraries, code generators, programming tools, static-analysis tools, build variants, configuration sources, and image formats. Identify which outputs are generated, how they are reproduced, and which tool qualifications or validations apply.

A released firmware image shall be traceable to source revision, configuration, toolchain, dependencies, hardware compatibility, build procedure, image hash, and approval evidence. Changes to silicon revision, board revision, compiler, optimization, RTOS, BSP, HAL, linker script, or generated-code input shall trigger an impact assessment.

B.12 Embedded Verification and Validation

Select test environments according to the requirement and the confidence needed. The verification plan may include review, static analysis, host unit tests, software-in-the-loop (SIL), processor-in-the-loop (PIL), target tests, hardware-in-the-loop (HIL), fault injection, timing and worst-case execution analysis, stress, endurance, power, thermal, environmental, electromagnetic, hardware-revision compatibility, boot, update, rollback, and recovery testing.

Each result shall identify the hardware revision, firmware version, build configuration, tool and fixture versions, calibration state, environmental conditions, input data, procedure, expected result, actual result, retained evidence, and reviewer. Simulation or host-test evidence shall not be represented as target evidence. Validation shall use representative intended operation and stakeholders or authorized representatives.

B.13 AI Tailoring Instructions

When this profile is Applicable or TBD, an AI agent shall read this appendix with the task-specific core chapters. Before editing, it shall ask for or confirm:

- the platform class: bare metal, RTOS, embedded Linux, programmable logic, or mixed;
- processor, board, programmable-device, and hardware revisions in scope;
- boot chain, operating system or RTOS, BSP, HAL, SDK, compiler, linker, and relevant third-party components;

- external interfaces, sensors, actuators, physical environment, timing, resource, power, safety, security, regulatory, manufacturing, update, and service constraints; and
- available schematics, datasheets, interface-control documents, hazard analysis, threat analysis, test evidence, and authorized approvals.

The agent shall add embedded concerns to the initialization gap list and place project facts only in their authoritative core chapters. It shall not invent pinouts, register behavior, electrical limits, memory sizes, execution times, power measurements, hardware compatibility, safety classifications, security claims, certification evidence, or test results. It shall report missing controlled hardware evidence and every decision requiring authorized human approval.

Use the following request for an embedded project:

```
Read tgmAIAgentGuide.tex and tgmEmbeddedSystemsProfile.tex. Apply the
Embedded Systems Tailoring Profile to [Project Name]. Ask me to
confirm the project namespace, profile applicability, platform class,
hardware revisions, toolchain, interfaces, constraints, applicable
standards, and available evidence. Add embedded concerns to the gap
list before editing. After human review, place project-specific
requirements, architecture, diagrams, tests, and evidence only in
their authoritative core chapters. Do not invent hardware facts or
verification results. Build the complete document, inspect the log,
and visually inspect the affected PDF pages.
```

B.14 Embedded-Profile Completion Checklist

Before completing embedded tailoring, confirm that:

- profile applicability and rationale are recorded;
- the hardware–software boundary and function allocation are explicit;
- timing, resources, power, interfaces, boot, recovery, safety, security, environmental, update, manufacturing, and lifecycle concerns were assessed;
- required embedded views and resource budgets exist or have documented omission rationales;
- hardware revisions, firmware versions, toolchain, build variants, and released images are configuration controlled;
- requirements, use cases, tests, traceability, verification, validation, and release records have reciprocal links; and
- the build succeeds and affected tables, diagrams, links, and page breaks have been visually inspected.

Appendix C

AI Agent Usage Guide

C.1 How to Use This Appendix

This appendix is the entry point for an AI agent that creates, reviews, or maintains this architecture specification. The agent shall read this file first, identify the requested task in the task-routing table, and load only the authoritative sources listed for that task. It shall expand the context when a referenced dependency, cross-cutting decision, or inconsistency requires it.

This appendix replaces substantive AI-oriented Markdown instructions. A thin repository bootstrap may point an AI tool to this appendix, but its routing tables, context packs, workflows, prompts, and checklists are maintained only as LaTeX so they remain visible in the architecture book and versioned with the specification.

C.2 Governing Principles

The authored LaTeX files and editable diagram sources are the source of truth. TikZ is the required source format for diagrams. The generated PDF is a review artifact and shall not be edited directly. The agent shall preserve one authoritative location for each fact and use links instead of duplicating content across chapters.

An AI agent is a drafting, analysis, consistency-checking, and build assistant. It shall use supplied project evidence, identify missing information, expose material assumptions, and distinguish proposals from approved decisions. It shall not invent requirements approval, implementation status, test results, verification evidence, validation evidence, waivers, or release authorization.

C.3 AI Quick Start

1. Read this appendix and identify the requested outcome.
2. Before creating or renaming a project-owned file, ask for or confirm the project file namespace. Do not invent it; use the confirmed namespace for every file created during

the task.

3. Inspect the domain-profile applicability table and the project profile decisions. Ask an authorized human to resolve any unreviewed profile selection. Load every Applicable or TBD profile affected by the task.

4. Select the closest entry in the task-routing table.

5. Read the listed primary files and the applicable context pack. Read `tgmManual.tex` when document metadata, macros, inclusion order, or build structure is affected.

6. Inspect available project evidence and report conflicts, missing facts, and consequential assumptions. Ask for a human decision when an answer would materially change scope, architecture, safety, security, cost, schedule, or release readiness.

7. Make narrowly scoped changes in the authoritative files. Preserve stable IDs and update every affected reciprocal mapping.

8. Add a linked document-history entry for an approved document change and record material AI interactions in Chapter D.

9. Run the complete build, inspect the log, render affected PDF pages, and complete the completion checklist.

C.4 Repository AI Bootstrap

The LaTeX architecture is authoritative, but an AI tool may require a repository-level discovery file before it knows where to find these instructions. Create a thin bootstrap file for each AI tool that the project actually uses. The bootstrap shall point to this chapter and the relevant LaTeX sources; it shall not reproduce their requirements, workflows, or architectural decisions. When policy changes, update the authoritative LaTeX source rather than copying the change into each bootstrap file.

Tool-mandated discovery names are documented exceptions to the project file namespace requirement. This exception applies only to the exact filename or location required for discovery. It does not exempt files created while performing the instructed work from the namespace requirements in the document-authoring requirements.

Create only the applicable files:

Table C.1: Repository AI Bootstrap Files

File	When to Create It
`AGENTS.md`	The repository is maintained with Codex or another agent that discovers this filename.
`CLAUDE.md`	The repository is maintained with Claude Code.
`.github/copilot-instructions.md`	The repository uses repository-wide GitHub Copilot instructions.

Repository AI Bootstrap Files (continued)

File	When to Create It
`SKILL.md`	The repository is intentionally packaged as, or contains, a reusable AI skill. Do not create this file merely as a second copy of `AGENTS.md`.

Each bootstrap file shall identify the root architecture source, point to this chapter, require task-appropriate context loading through the task-routing table, direct architecture policy changes back to LaTeX, and point to Chapter A for building and verification. It shall also state the confirmed project file namespace or direct the agent to obtain it before creating project-owned files. Use the following content as a minimal tool-neutral pattern, adapting only the discovery-specific wrapper:

```
# AI Instructions

The authoritative software architecture is maintained in
tgmManual.tex. Before architecture-related work, read
tgmAIAgentGuide.tex and use its task-routing table to identify the
additional LaTeX chapters required for the task.

Treat the authored LaTeX and editable diagram sources as the source
of truth. Do not duplicate their requirements, conventions,
workflows, or architectural decisions in this file. When policy
changes, update the authoritative LaTeX source.

Before creating project-owned files, obtain or confirm the project
namespace required by tgmDocumentGuide.tex. Follow that guide's
build and verification instructions before completing the task.
```

C.5 Task Routing

The file lists below are minimum starting contexts, not permission to ignore a dependency discovered during the task. A task that changes an artifact shall also inspect every register that contains a reciprocal mapping to that artifact.

Table C.2: AI Task-Routing Table

Task	Minimum Authoritative Sources
Initialize a project	`tgmManual.tex`, `prologue.tex`, `tgmIntroduction.tex`, `tgmDevelopmentPlan.tex`, `tgmSystemContext.tex`, each proposed or selected domain profile, and the supplied project evidence.

Continued on next page

AI Task-Routing Table (continued)

Task	Minimum Authoritative Sources
Tailor an embedded system	`tgmEmbeddedSystemsProfile.tex`, `tgmDevelopmentPlan.tex`, `tgmSystemContext.tex`, `tgmRequirements.tex`, the affected architecture and interface chapters, `tgmTests.tex`, `tgmTraceability.tex`, and the controlled hardware and project evidence.
Create repository AI bootstrap files	`tgmAIAgentGuide.tex`, `tgmDocumentGuide.tex`, `tgmManual.tex`, evidence identifying the AI tools used by the project, and each tool's required discovery filename.
Develop user stories	`tgmSystemContext.tex`, `tgmUserStories.tex`, `tgmRequirements.tex`, and stakeholder evidence.
Add or change requirements	`tgmUserStories.tex`, `tgmRequirements.tex`, every Applicable or TBD domain profile, `tgmUseCases.tex`, `tgmTests.tex`, `tgmTraceability.tex`, `tgmVerification.tex`, and `tgmGlossary.tex`.
Add or change use cases	`tgmUserStories.tex`, `tgmRequirements.tex`, `tgmUseCases.tex`, `tgmTests.tex`, and `tgmTraceability.tex`.
Develop or change architecture	`tgmSystemContext.tex`, `tgmRequirements.tex`, `tgmArchitectureOverview.tex`, the affected interface, logical, process, development, physical, or implementation chapters, and their `TikZ/` diagram sources.
Add or change tests	`tgmRequirements.tex`, `tgmUseCases.tex`, `tgmTests.tex`, `tgmTraceability.tex`, `tgmVerification.tex`, and `tgmTestEvidence.tex`.
Perform verification	`tgmRequirements.tex`, `tgmTests.tex`, `tgmTraceability.tex`, `tgmVerification.tex`, and controlled test evidence.
Perform validation	`tgmUserStories.tex`, `tgmUseCases.tex`, `tgmValidation.tex`, and controlled stakeholder evidence.
Prepare a release	`tgmRequirements.tex`, `tgmTraceability.tex`, `tgmVerification.tex`, `tgmValidation.tex`, `tgmRelease.tex`, and controlled release evidence.
Review consistency	All registers containing the affected artifact types, `tgmGlossary.tex`, `tgmDocumentHistory.tex`, and `tgmManual.tex`.
Compile or troubleshoot	`tgmManual.tex`, `tgmDocumentGuide.tex`, the affected chapter, its diagram sources, and the build log.

C.6 Task-Specific Context Packs

A context pack is a reading boundary within this appendix, not a separate file. Use the applicable pack together with the task-routing entry.

C.6.1 Domain-Profile Context

Determine the Applicable, Not Applicable, or TBD status of every available domain profile during initialization. An authorized human shall approve the selection and rationale. For later tasks, load every Applicable or TBD profile whose concerns intersect the requested change. A profile adds review concerns and artifact expectations; project-specific facts remain in their authoritative core chapters. When the Embedded Systems profile applies, follow Chapter B and add hardware, firmware, timing, resource, power, physical-interface, safety, security, manufacturing, update, and field-service concerns to the task context. For requirements work, apply the controlled translation rule in Section 5.6: each applicable profile concern shall map to a requirement, an existing requirement, or an approved Not Applicable decision with rationale.

C.6.2 Requirements Context

Establish the system boundary and stakeholder need before proposing a requirement. Keep each requirement singular, feasible, unambiguous, and verifiable. Apply the identifier grammar and family definitions in Section 5.2. Classify its primary family or subtype, write binding behavior with **shall**, assign one Must, Should, Could, or Would priority, assign one lifecycle status, and map it to at least one use case and one test case. A requirement may be marked Done only when its mapped tests pass, verification is complete, traceability is current, and a controlled implementation version is known. Evaluate every Applicable or TBD domain profile that intersects the proposed change and follow Section 5.6 before completing the requirements work.

C.6.3 Use-Case Context

Start from an actor goal and a system boundary. Give each use case its own table, identify participating actors, preconditions, trigger, success outcome, main flow, alternate and exception flows, postconditions, mapped requirements, and mapped tests. Describe externally observable interaction rather than internal implementation detail. Maintain reciprocal mappings to every related requirement.

C.6.4 Architecture Context

Trace each architectural decision to requirements, quality attributes, constraints, risks, and interfaces. Identify alternatives and explain the trade-off that selected the proposed structure. Keep the context, logical, process, development, physical, interface, and implementation views consistent. Use TikZ for diagrams. Store each TikZ source under its owning chapter as required by Section A.7, include it from that chapter, and compile the PDF.

C.6.5 Testing and Evidence Context

Derive tests from requirements and use cases. Define preconditions, inputs, procedure, expected result, and retained evidence. A test definition is not a test result. Verification records

objective compliance with requirements; validation records whether stakeholder needs are satisfied in intended use. Never create passing evidence or approval from an expected result.

C.6.6 Release Context

Review open, in-progress, done, and retired requirements; unresolved defects; verification and validation results; operational readiness; migration and rollback provisions; known limitations; and required approvals. Report a gap instead of treating missing evidence as acceptance. Only an authorized human may approve release.

C.6.7 Document-Authoring Context

Follow Chapter A for file naming, LaTeX conventions, diagrams, installation, and builds. Update the glossary and index when introducing a controlled term. Update document history with a link to the changed location. Before creating or renaming a project-owned file, ask for or confirm the project file namespace and apply it consistently in accordance with the document-authoring requirements. Every TikZ source shall satisfy the path, inclusion, caption, label, and reference requirements in Section A.7. Do not edit generated auxiliary files or the PDF directly.

C.7 Mandatory Consistency Rules

1. Preserve every stable ID. Do not reuse an ID belonging to a retired or deleted proposal.
2. Use the scope and family codes in Section 5.2. Before using an undefined scope or family, propose its code, name, boundary, rationale, and examples; check for semantic overlap and identifier collision; obtain authorized-human approval; and update the convention, category catalog, this guide, and the applicable register. Do not rename existing IDs when extending the convention.
3. Use **shall** for binding requirements. Use priority terms only in the Priority field and with the definitions in the glossary.
4. Give every requirement a priority, status, In Version value, use-case mapping, and test-case mapping.
5. Enter ∅ when a requirement is not Done and no completed implementation version applies.
6. Maintain reciprocal mappings among user stories, requirements, use cases, tests, traceability, verification, and validation.
7. Do not copy one controlled statement into several authoritative locations. Link to its stable ID.

8. Retain retired requirements and their history. Record the rationale and replacement when applicable.

9. Use project evidence for factual claims. Label an unsupported suggestion as a proposal and report the evidence needed to decide it.

10. Record each domain profile as Applicable, Not Applicable, or TBD with an authorized decision and rationale. Load every Applicable or relevant TBD profile; do not duplicate project facts in the profile appendix.

11. Before creating or renaming any project-owned file, ask for or confirm the project `<nameSpace>`; do not invent it. Prefix every new project-owned file name with the confirmed namespace.

12. Use TikZ for every diagram. Every TikZ source shall use the directory

    `TikZ/<includingFileBaseName>/`

    and shall be named `<nameSpace><DrawingName>.tex`. The directory shall exactly match the base name of the LaTeX file containing its `\input` command. The including chapter shall place it in a numbered figure, add a descriptive caption and stable label, and reference the figure in its text.

13. Link every document-history entry to the location where the change occurred.

14. Compile after structural, cross-reference, glossary, index, diagram, or table changes and visually inspect the affected pages.

C.8 Standard AI Workflows

C.8.1 Initialize a Project

Ask an authorized human for the project file namespace, or confirm an already configured namespace, before creating or renaming files. Do not invent it. Replace `\FileNamespace`, project metadata, and bracketed placeholders only from supplied facts, then apply the namespace to every new project-owned file. Identify available domain profiles and ask an authorized human to confirm each profile's applicability. Record the decisions in the project profile table. Include each Applicable or TBD profile in the initialization context; remove an inapplicable profile appendix only after its decision is recorded. Create a gap list covering purpose, scope, stakeholders, system boundary, interfaces, constraints, quality attributes, risks, evidence, and approvals. Resolve consequential gaps before developing detailed artifacts. Work in dependency order from context and needs through requirements, use cases, architecture, tests, evidence, and release.

C.8.2 Add or Change a Requirement

Identify the originating need and affected use cases. Draft a singular verifiable obligation, select its scope, primary family, priority, and status, and assign a stable ID using Section 5.2. If no defined scope or family fits, complete the proposal and human-approval procedure before assigning the ID. Add or update mapped use cases and tests, then update traceability and verification. Confirm every link in both directions and add a linked history entry.

C.8.3 Add or Change a Use Case

Identify the actor goal, system boundary, and mapped requirements. Complete one use-case table with success, alternate, and exception behavior. Add mapped tests, update requirement mappings and traceability, and confirm that no flow introduces an unrecorded requirement.

C.8.4 Develop or Change Architecture

Identify the requirements and quality attributes that drive the decision. Describe alternatives, trade-offs, risks, interfaces, and the selected structure. Update every affected architectural view and diagram. Confirm that the proposed design neither omits a driving requirement nor silently adds an unapproved constraint.

C.8.5 Add Tests or Evidence

Derive each test from mapped requirements and use-case flows. Define repeatable steps and objective expected results. Link retained evidence only after an actual execution. Update verification or validation records without converting an expected outcome into a claimed result.

C.8.6 Review and Prepare a Release

Audit IDs, mappings, statuses, versions, unresolved assumptions, verification, validation, operational readiness, and release criteria. Report contradictions and missing evidence. Prepare a release recommendation for human approval; do not approve the release.

C.9 Example AI Requests and Prompts

Replace bracketed text with project facts. A prompt does not override this guide, the controlled project evidence, or human approval boundaries.

C.9.1 Initialize the Architecture Book

```
Read tgmAIAgentGuide.tex and use the Initialize a Project workflow
for [Project Name]. Review the supplied project evidence and create
a gap list before editing. Identify missing scope, stakeholders,
```

```
interfaces, constraints, quality attributes, risks, evidence, and
approvals. Identify the available domain profiles and ask me to
confirm whether each is Applicable, Not Applicable, or TBD. Read each
Applicable or relevant TBD profile and add its concerns to the gap
list. After the gap list is reviewed, replace only supported
placeholders and develop the document in dependency order. Compile
the complete PDF and report unresolved human decisions.
```

C.9.2 Tailor an Embedded System

```
Read tgmAIAgentGuide.tex and tgmEmbeddedSystemsProfile.tex. Apply the
Embedded Systems Tailoring Profile to [Project Name]. Ask me to
confirm the project namespace, profile applicability, platform class,
hardware revisions, toolchain, interfaces, constraints, applicable
standards, and available evidence. Add embedded concerns to the gap
list before editing. After human review, place project-specific
requirements, architecture, diagrams, tests, and evidence only in
their authoritative core chapters. Do not invent hardware facts or
verification results. Build the complete document, inspect the log,
and visually inspect the affected PDF pages.
```

C.9.3 Create Repository AI Bootstrap Files

```
Read the Repository AI Bootstrap section in tgmAIAgentGuide.tex and
treat that LaTeX chapter as authoritative. Confirm which AI tools
this repository uses and the project file namespace. Create only the
applicable repository discovery files, such as AGENTS.md or
CLAUDE.md. Create SKILL.md only when the repository is intentionally
packaged as, or contains, a reusable skill. Keep every bootstrap file
short: point to tgmManual.tex, tgmAIAgentGuide.tex, the task-routing
table, and tgmDocumentGuide.tex. Do not duplicate architecture rules.
Verify that each file directs architecture-related changes back to
the authoritative LaTeX sources.
```

C.9.4 Develop Requirements

```
Read tgmAIAgentGuide.tex and follow the Requirements context and
workflow. Load every Applicable or TBD domain profile that intersects
the work. From the approved objectives and user stories, propose
singular, feasible, and verifiable requirements. Classify each one,
use shall for the binding obligation, assign one priority and one
status, and map it to at least one use case and test case. Do not
complete requirements work until each applicable profile concern maps
to a new or existing requirement, or to an approved Not Applicable
```

```
decision with rationale. Do not mark work Done or assign an
implementation version without evidence.
Update reciprocal mappings, history, and the index; then build and
inspect the affected pages.
```

C.9.5 Develop Use Cases

```
Read tgmAIAgentGuide.tex and follow the Use-Case context and
workflow. Derive use cases from the approved actor goals and mapped
requirements. Give each use case its own table with actors, trigger,
preconditions, main flow, alternate and exception flows, outcome,
postconditions, requirements, and tests. Maintain reciprocal links
and report any flow that requires a new requirement. Build and
inspect the affected pages.
```

C.9.6 Develop an Architecture Decision

```
Read tgmAIAgentGuide.tex and follow the Architecture context and
workflow for [decision or subsystem]. Identify driving requirements,
quality attributes, constraints, interfaces, and risks. Compare
reasonable alternatives and document the trade-off supporting the
proposed choice. Update every affected architecture view and editable
diagram source. Do not present the proposal as approved. Compile the
document and inspect all affected diagrams and pages.
```

C.9.7 Develop Tests and Traceability

```
Read tgmAIAgentGuide.tex and follow the Testing and Evidence context.
For [requirements or use cases], define repeatable tests with inputs,
preconditions, steps, expected results, and required evidence. Update
all reciprocal mappings and the traceability matrix. Do not invent an
execution result or mark verification complete. Compile the document,
check references, and inspect the affected tables.
```

C.9.8 Review Release Readiness

```
Read tgmAIAgentGuide.tex and follow the Release context and workflow
for [release]. Audit requirement status and implementation versions,
traceability, verification, validation, defects, operations, rollback,
known limitations, and approvals. Separate confirmed evidence from
gaps and assumptions. Prepare a release-readiness recommendation for
human review, but do not approve release. Build and inspect the final
document.
```

C.10 Human Decisions and Information Boundaries

An authorized human shall approve scope, requirements, architecture decisions, security and privacy posture, safety decisions, evidence sufficiency, waivers, validation conclusions, and release. The agent shall present alternatives and evidence without claiming that review has occurred.

Do not provide an AI agent with credentials, secrets, personal data, production data, export-controlled material, or other information it is not authorized to process. Before using external sources, confirm that the task permits network access and that the source may be incorporated. Record material prompts, assumptions, corrections, and unresolved issues in Chapter D.

C.11 Completion Checklist

Table C.3: AI-Assisted Change Completion Checklist

Area	**Required Completion Evidence**
Scope	The requested outcome is complete; unrelated chapters and user changes remain untouched.
Domain profiles	Every available profile has an authorized applicability decision and rationale; every Applicable or relevant TBD profile was included in the task context, and project facts remain in core chapters.
Evidence	New factual statements are supported by supplied or controlled evidence; proposals, assumptions, and unknowns are identified.
Language	Binding requirements use shall; priority and status use the controlled terms and definitions.
Identifiers	Stable IDs follow the documented scope and family convention, remain unique, and are neither renumbered nor reused; every new scope or family has its definition and authorized-human approval recorded before use.
File names	The project namespace was supplied or explicitly confirmed; every newly created project-owned file begins with it, and no namespace was invented.
AI bootstrap	Every applicable repository AI discovery file is a thin pointer to the authoritative LaTeX architecture; tool-mandated filenames are documented exceptions, and no architecture rules are duplicated in Markdown.
Traceability	Every changed requirement, use case, test, verification record, and validation record has complete reciprocal mappings.
History	The newest approved document change appears at the top of document history and links to the changed location.
Glossary and index	New controlled terms and important concepts are defined and indexed where applicable.
Diagram source	Every diagram uses TikZ; each TikZ directory exactly matches its including LaTeX file's base name, each drawing file begins with the project namespace, and the drawing is included, captioned, labeled, and referenced.
Build	The complete platform build succeeds; citations, glossary, index, diagrams, and cross-references are regenerated.
Log review	No fatal error, undefined reference, duplicate label, missing file, or unresolved layout warning remains.
Visual review	Affected PDF pages show readable tables, correct page breaks, current diagrams, working headings, and no clipped or overlapping content.
Human review	Decisions requiring authority are clearly identified and are not represented as approved without evidence.

Appendix D

AI Prompt Log

D.1 How to Use This Appendix

Use this log when AI assistance materially affects requirements, architecture, code, tests, or release evidence. Record enough context for a reviewer to understand the input, output, assumptions, and human corrections.

D.2 Example Record

Date: *[timestamp]*; Tool and model: *[name]*; Input: *[requirements and constraints supplied]*; Task: *[requested output]*; Human review and corrections: *[summary]*; Unresolved assumptions: *[list or none]*.

D.3 Requirement Scope or Family Extension Record

When an AI-assisted change proposes a requirement scope or family that is not already defined, record the proposed uppercase code, full name, decision boundary, rationale, representative examples, semantic-overlap review, identifier-collision check, affected register and category-catalog changes, and the authorized human's approval or rejection. Do not assign an identifier from the proposed scope or family before approval.

D.4 Project Initialization Record

Date: October 14, 2026; Tool and model: Anthropic Claude Code (Claude Opus 5.5); Input: the template initialization request in Section 1.7, the project name toggle-manhattan, file namespace `tgm`, AI tool selection (Claude Code only; no reusable skill), Embedded Systems profile marked Applicable, document metadata, and the statement that the system controls the opening and closing of a curtain through an undecided method (physical controller or phone application); Task: produce an initialization gap list, then initialize the project copy, create applicable AI bootstrap files, and build and inspect the document; Human review

and corrections: the requester confirmed the namespace, AI tools, profile applicability, and metadata, and stated that no further project evidence exists; Unresolved assumptions: none adopted. Open decisions are listed in the open context items; the authorized approver and the embedded-profile decision owner were initially TBD. Follow-up decisions on October 14, 2026: Stel Angelidis is the authorized approver and approved the Embedded Systems profile decision; the control method is both a physical controller and a phone application; the embedded platform class remains TBD; Claude Code history entries use the initials CLAUDE; the stakeholders are people who want a low-cost, non-privacy-invading IoT curtain closer; privacy means no data collection about individual people and no cameras or microphones; components cost no more than USD 20 (cost basis TBD); the system runs on a microcontroller, with all other platform details TBD. The template's example requirements, use cases, tests, and registers remain unchanged as generic examples and are not project content.

Appendix E

Test Evidence

E.1 How to Use This Appendix

Store or reference detailed test and verification evidence here when it would make the main chapters difficult to read. Evidence must identify the artifact, result, environment, date, and product revision.

E.2 Example Evidence Record

Test: *TC-001*; Result: *Pass*; Environment: *[browser, operating system, or deployment]*; Revision: *[build ID]*; Evidence: *[link to report, log, screenshot, or automated result]*. Redact personal or production-sensitive data.

Glossary

acceptance criterion An observable condition used to decide whether a need, story, or requirement has been satisfied. 75

Could A desirable capability included only when time and resources permit. 2, 21, 64, 75

Must A capability required in the current release; the release is unacceptable without it. 2, 10, 20–23, 64, 75

requirement A necessary, singular, unambiguous, feasible, and verifiable statement of capability or constraint. 75

Should An important capability expected in the current release unless a documented constraint prevents it. 2, 10, 21, 22, 64, 75

stakeholder A person, group, or organization that affects the system or is affected by it. 75

traceability The ability to follow an artifact backward to its source and forward to its implementation and evidence. 75

use case A goal-oriented interaction between an actor and the system, including normal and alternate flows. 75

V-model A lifecycle model that pairs each decomposition and specification activity with a corresponding integration and evidence activity. 4, 75

validation Confirmation that the completed system satisfies stakeholder needs in its intended context; building the right system. 75

verification Confirmation that specified requirements have been fulfilled; building the system right. 75

Would A capability assigned to a later implementation version after higher-priority work. 2, 10, 21, 64, 75

Acronyms

AI artificial intelligence. 1, 75

IoT Internet of Things. 8, 75

UI user interface. 75

UML Unified Modeling Language. 75

V&V verification and validation. 75

Index

acceptance criterion, 10
AGENTS.md, 61
AI agent, 60
  architecture workflow, 4, 44, 66
  completion checklist, 70
  consistency rules, 65
  context pack, 63
  context routing, 60
  embedded systems tailoring, 58
  example prompts, 60, 67
  human review, 60, 70
  LaTeX installation prompt, 44
  project initialization, 2
  quick start, 60
  repository bootstrap, 61
  task routing, 62
  TikZ requirement, 60
Appendix
  AI Agent Usage Guide, 60
  Document Authoring Guide, 44

build tools, 44

Chapter
  Introduction, 1
CLAUDE.md, 61
cost limit, 8

diagram
  TikZ source, 46
document conventions, 44
document history, v
domain tailoring profile, 3
  embedded systems, 52

embedded systems, 52

hardware–software co-design, 52

introduction, 1

LaTeX, 44

microcontroller, 8
Muresan, Darian, iii

normative language, 1

priority, 1, 12
privacy, 8

Requirement
  REQ-B-001, 20
  REQ-C-001, 23
  REQ-D-001, 23
  REQ-F-001, 21
  REQ-F-002, 21
  REQ-F-003, 21
  REQ-I-001, 22
  REQ-Q-001, 21
  REQ-S-001, 22
requirement, 12
  business, 20
  category catalog, 15
  constraint, 23
  data, 23
  domain profile, 15
  family code, 12
  functional, 14
  identifier convention, 12
  implementation version, 12
  interface, 22
  quality, 14
  scope qualifier, 12
  security, 22
  status, 14
requirements traceability matrix, 36

SKILL.md, 61
source of truth, 1, 60
stakeholder, 8
status
    Done, 14
    In Progress, 14
    Open, 14
    Retired, 14

template
    project initialization, 2
    tailoring, 3
template attribution, iii
template provenance, 44
TestCase
    TC-001, 34
    TC-002, 34
    TC-003, 34
    TC-004, 34
    TC-005, 34
    TC-006, 35
    TC-007, 35
    TC-008, 35
    TC-009, 35
TikZ, 44
traceability, 36

use case diagram, 24
UseCase
    UC-001, 24
    UC-002, 25
    UC-003, 25
user story, 10
UserStory
    US-001, 10
    US-002, 10
    US-003, 10

V-model, 1
Validation
    VAL-001, 40
    VAL-002, 40
    VAL-003, 41
validation, 40
validation scenario, 40
Verification
    VER-001, 38
    VER-002, 38
    VER-003, 38
    VER-004, 38
    VER-005, 38
    VER-006, 38
    VER-007, 38
    VER-008, 39
    VER-009, 39
verification, 38
verification method, 38